

Targeting Persistent Disagreement: Coarse Interventions in Social Networks*

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July 2026

Abstract

Institutions that shape public opinion often change intervention intensity more easily than whom they reach. We study a decision-maker who commits to one exposure rule and reuses it over time. This restriction is vacuous in one period but consequential dynamically, because the rule changes the path on which it will later be used. Under equal centrality, optimal exposure follows disagreement that persists, not disagreement that is merely large today. Near a symmetric benchmark, reuse makes the tilt toward persistent disagreement smaller than in the one-period problem. When centrality differs, exposure also follows leverage: an agent’s propagated influence over future opinions. The value of retargeting depends on the realized path: one fixed rule can perform nearly as well as unrestricted targeting when that path is low-dimensional and the two policies agree on its leading directions. Using Yale climate opinions and Facebook social connectedness, we find that California has greater authority than Texas but nearly the same exposure, consistent with Texas’s greater future persistence; reuse also materially reallocates exposure. At the same time, the realized path is nearly one-dimensional, so one fixed rule captures nearly all attainable gains.

JEL: D85, D83, C63.

Keywords: Coarse targeting, Persistent disagreement, Network leverage, Dynamic intervention, Opinion dynamics.

*This paper supersedes an earlier version circulated under the title “Coarse Targeting in Social Networks.” We thank Ying Gao, Ben Golub, Li, Hao, Matt Jackson, Jesse Perla, Mike Peters, Amin Rahimian, Marit Rehavi, Euncheol Shin, Kyungchul (Kevin) Song, Wing Suen, Xu Tan, Philip Ushchev, Yiqing Xing and Junjie Zhou for great comments. We also thank seminar participants at UBC, SMU, UW, and EC 2026 for feedback. This work was completed independently of Philip Solimine’s employment at Microsoft and does not reflect the views or positions of Microsoft Corporation or its affiliates. Email: wei.li@ubc.ca; philsolimine@microsoft.com.

1 Introduction

Institutions that seek to shape public opinion redesign whom they reach less often than how intensely they intervene. Health agencies, political campaigns, and social-media platforms operate through exposure rules – which regions receive more advertising, which communities see more prominent messages, which content is amplified – that fix the relative pattern of exposure while overall intensity adjusts with budgets, enforcement effort, or ranking salience. Meta’s third-party fact-checking program illustrates the distinction: changing the organizations it worked with and the languages it covered required organizational, policy, and compliance decisions, while the intensity with which flagged content was down-ranked or labelled could adjust more readily (Facebook, 2016; Meta, 2025b,a; Gorwa, 2019; Krishnan et al., 2021).

We study this asymmetry between exposure pattern and intensity as *coarse targeting* in an opinion network. The decision-maker commits at the outset to an exposure vector $\mathbf{b} \in \mathbb{R}^n$ and then chooses a scalar intensity r_t in each period. This is a problem of instrument simplicity: one exposure pattern replaces a multidimensional intervention that could otherwise be redesigned each period. In a one-period problem, the restriction is vacuous—any vector intervention $\mathbf{u}_0$ can be written as $\mathbf{b}r_0$. Its entire content is intertemporal: the same rule must be reused.

Reuse makes the problem endogenous. The rule shapes the opinion path on which it will later be used, so it must be chosen against the path it creates. Two questions follow. What should a reusable rule target? And what is targeting flexibility worth? More generally, the problem is how to design a simple instrument when its value depends on the path that the instrument itself creates.

The first question is not answered by the most extreme opinions today or by centrality alone. Consider four agents divided into two symmetric islands, $\{1, 2\}$ and $\{3, 4\}$. Each agent places weight 0.4 on each opinion in her own island and 0.1 on each opinion in the other island, so all four agents have equal centrality. Suppose

$$\mathbf{x}_0 = (100, -99, 1, 1)'.$$

Agents 1 and 2 differ by 199, while the average opinions of the two islands differ by only 0.5. Social updating erases the within-island gap after one update, and optimal exposure

therefore assigns agents 1 and 2 identical exposure, regardless of how far apart their opinions are. What survives is the much smaller disagreement between islands: the extreme opinions in island 1 largely offset each other, while the common opinion in island 2 remains part of the path the network carries forward. Optimal targeting tilts toward island 2. A reusable rule targets disagreement according to whether the network preserves it, not according to its current magnitude.

The example isolates the first force: persistence. More generally, when agents are equally central – even in a directed, asymmetric network – uniform exposure is optimal exactly when updating eliminates inherited disagreement. If disagreement survives, exposure responds to what the network carries forward; for positively persistent dimensions, the response grows with persistence. Reuse dampens this response: near a symmetric benchmark, a reusable rule tilts less toward persistent disagreement than a one-period rule, because the same exposure pattern remains in place after that disagreement has decayed. In the two-island example, the island-2 to island-1 exposure ratio is 1.50 in the one-period problem but only 1.15 under reuse.

When centrality differs, a second force appears: leverage. Near the same benchmark, a redistribution of authority propagates along network paths in Katz–Bonacich form, and optimal exposure tilts toward this effective influence rather than the raw shift. In the opinion-leader benchmark, greater authority attracts more exposure, but concentrating exposure on one leader creates temporary disagreement, so the optimal rule stays dispersed. Persistence determines what remains worth moving; leverage determines who can move it.

Under full targeting, as in Li and Tan (2026), the decision-maker chooses an unrestricted intervention vector $\mathbf{u}_t$ each period rather than requiring $\mathbf{u}_t = \mathbf{b}r_t$. The value of flexibility depends jointly on where the endogenous opinion path places weight and on where a fixed rule differs from unrestricted targeting. Formally, the value gap is exactly the trace of a product of two matrices representing these two objects. A fixed rule can therefore attain the full-targeting value even if the policies differ away from the realized path; equivalence obtains exactly when they choose the same intervention at every date along the path induced by the fixed rule. More generally, the gap is small when path weight is concentrated in a few directions and the policies agree on them. We call the number of directions carrying substantial path weight the *effective path dimension*. The cost of using a simple instrument therefore depends on the path’s effective dimension and on policy agreement along its

leading directions, not on the number of agents or the rank of the network. The directions a simple instrument must span are determined by the path on which it will be used, not by the dimension of the environment it operates in.

We illustrate both questions using U.S. state-level climate opinions and the Facebook Social Connectedness Index. The network supplies the propagation matrix, while the climate data provide initial disagreement across states. California has about 21% more authority weight than Texas, yet receives only about 1% more exposure under the highest-value coarse rule we compute. The state comparison is consistent with Texas’s higher future-persistence score offsetting most of California’s authority advantage. Ohio shows the reverse, receiving more exposure than its authority alone would predict.

The application also answers the second question. Even after removing the common opinion component, 96.2% of discounted path weight lies in one direction, and the computed coarse rule captures 99.6% of attainable gains. One well-chosen fixed rule can thus outperform fixed rules based on centrality or current opinions and even period-by-period myopic retargeting. At the baseline, accounting for reuse substantially reallocates exposure (Section 6.2): California’s weight falls from 0.437 under the one-period rule to 0.328 under the reusable rule, while Texas’s falls from 0.398 to 0.325. Thus reuse can materially change whom the decision-maker should target even when the value of subsequent retargeting is small.

The boundary is not whether disagreement keeps arriving, but whether it arrives in many new directions. Recurrent agent-level shocks inject new disagreement in unpredictable directions and can eventually make flexibility decisive. A persistent bias repeatedly injects the same source and, in the highly symmetric benchmark, adds at most one relevant disagreement direction. Recurrence alone does not defeat a reusable rule; diffuse new disagreement does.

Related literature

A. Network intervention and targeting. A large literature studies how outcomes in networks respond to targeted interventions. Early work emphasizes network position, centrality, and key-player measures (Galeotti and Goyal, 2009; Ballester et al., 2006); more recent work analyzes optimal intervention when agents interact strategically or spillovers are present (Candogan et al., 2022). Closest in method is Galeotti et al. (2020), who decompose a one-shot intervention into the principal components of a static network game. The spectral logic is related, but reuse changes the relevant object: persistence determines which opinion

components remain valuable, and their weights are generated by the path the exposure rule itself induces rather than by a fixed network matrix alone. Relatedly, Akbarpour et al. (2025) show that the value of network-optimal seeding can be limited in diffusion problems. Our low value of additional targeting flexibility arises instead from concentration of the endogenous opinion path.

In dynamic opinion models, interventions are typically tailored across agents or adjusted as opinions evolve (Ghaderi and Srikant, 2014). Closest among these is Li and Tan (2026), which shares the linear-updating environment but asks a different question: how competing influencers with opposing agendas choose interventions in equilibrium. Here a single decision-maker commits to one exposure direction and reuses it, so the object of interest is the restriction itself rather than strategic interaction.¹ Sadler (2023) studies how stochastic opinion fluctuations shape the concentration of influence campaigns in a network with stubborn agents.

B. Simple instruments and exposure design. Economic institutions often use uniform or low-dimensional instruments even when finer tailoring is feasible. Retail chains set nearly uniform prices across heterogeneous local markets (DellaVigna and Gentzkow, 2019); in mechanism design, a second-price auction with a common reserve is optimal when bidders’ values are i.i.d. and regular (Myerson, 1981) and retains approximation guarantees when bidders differ (Hartline and Roughgarden, 2009). Related work rationalizes simple contractual forms under robustness considerations (Carroll, 2015). Classical instrument-counting arguments compare the number of instruments with the number of independent policy targets (Tinbergen, 1952). Our restriction is intertemporal: the exposure pattern is chosen initially but must then be reused, so the relevant count is the number of intervention directions required along the endogenous opinion path.

In digital environments, limits on personalization arising from privacy, regulation, transparency, or reputational concerns similarly motivate reusable exposure rules (Goldfarb and Tucker, 2011; Lambrecht and Tucker, 2013; Choi et al., 2019). We characterize what such rules should target and when additional targeting flexibility is valuable.

C. Opinion dynamics. We use linear updating as a tractable representation of social interaction (DeGroot, 1974; Friedkin and Johnsen, 1990; DeMarzo et al., 2003; Golub and

¹Relatedly, Gaitonde et al. (2020) and Mostagir et al. (2022) examine how network structure shapes vulnerability to disagreement and misinformation.

Jackson, 2010). Our focus is not on whether opinions converge, but on how an institution should intervene when current exposure changes the future marginal value of influence. Models with behavioral or informational frictions, including locally Bayesian learning in Li and Tan (2020) and related work (Molavi et al., 2018; Li and Tan, 2021), often generate structured dynamics that can be approximated locally by linear updating rules. Our analysis applies to the local propagation structure of such richer environments.

D. Dynamic systems and intervention. In engineering and applied mathematics, a large literature studies how intervention changes the evolution of network outcomes, often emphasizing feasibility, controllability, or stabilization (Lin, 1974; Pasqualetti et al., 2014; Liu and Barabási, 2016). Some recent work compares or selects where intervention enters the network (Summers et al., 2016; Lindmark and Altafini, 2018, 2020; Gao et al., 2021; Klickstein and Sorrentino, 2023).

Our problem differs in both instrument and objective. The decision-maker reuses one exposure direction and minimizes losses along the entire opinion path, so the value of flexibility depends on where that path places weight and how closely the fixed rule matches unrestricted intervention there.

2 Model

A network is represented by a weighted directed graph $G = (N, A)$, where $N = \{1, \dots, n\}$ and $A = [a_{ij}]$ is row-stochastic, with $a_{ij} \geq 0$ and $\sum_j a_{ij} = 1$ for every i .² The entry a_{ij} is the weight agent i places on agent j 's opinion. A node may represent an individual, group, region, or other unit at which exposure is implemented. In Section 6, nodes are U.S. states.

Time is discrete. Let $\mathbf{x}_t \in \mathbb{R}^n$ denote opinions at date $t = 0, 1, 2, \dots$. After observing $\mathbf{x}_0$, a decision-maker commits at $t = 0$ to an exposure rule $\mathbf{b} \in \mathbb{R}^n$.³ In each period, she observes $\mathbf{x}_t$ and chooses a scalar intervention level $r_t \in \mathbb{R}$. Opinions evolve according to

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t.$$

²We use boldface lowercase letters for vectors and boldface uppercase letters for matrices, and all vectors are column vectors.

³We maintain that A and $\mathbf{x}_0$ are known when the exposure rule is chosen. Bloch and Shabayek (2023) study targeting under uncertainty about network position.

The rule $\mathbf{b}$ fixes the relative exposure pattern across agents, while r_t scales and, when negative, reverses that pattern. The total intervention is $\mathbf{u}_t = \mathbf{b}r_t$.⁴

Exposure is measured relative to a no-intervention baseline. For a positive intensity, positive coordinates of $\mathbf{b}$ raise exposure above baseline and negative coordinates lower it. Thus $\mathbf{b}$ is a relative exposure rule, not an allocation of a fixed positive budget.⁵ The decision-maker cannot change this rule across agents over time. This *coarse targeting* restriction is entirely intertemporal. Any single-period intervention $\mathbf{u}_0$ can be written as $\mathbf{b}r_0$ with $\|\mathbf{b}\| = 1$.

We impose $\|\mathbf{b}\| \leq 1$, a capacity bound. Increasing $\|\mathbf{b}\|$ permits the same intervention path to be implemented with smaller scalar intensities and hence at lower cost. Lemma 2 shows that the bound is attained whenever intervention is strictly valuable.⁶

The technology applies whenever an institution runs one relative exposure rule rather than several independently adjustable rules in parallel.

Platform ranking. Nodes are communities or content categories, and $\mathbf{b}$ fixes their relative amplification or down-ranking from baseline. Changing the pattern requires product, policy, or compliance review, while its overall salience r_t adjusts more frequently.

Regulatory attention. Nodes are jurisdictions or firms, and $\mathbf{b}$ specifies how enforcement or outreach is reallocated from baseline. The pattern changes slowly, while the overall intensity of inspections, audits, or outreach responds to current conditions.

Fix $\mathbf{b}$, a discount factor $\delta \in (0, 1)$, and a cost parameter $c > 0$. The decision-maker chooses $\{r_t\}_{t \geq 0}$ to solve

$$\max_{\{r_t\}_{t \geq 0}} - \sum_{t=0}^{\infty} \delta^t (\mathbf{x}'_t Q \mathbf{x}_t + cr_t^2)$$

where $\mathbf{x}'_t Q \mathbf{x}_t$ is the loss from opinion deviations and cr_t^2 is the cost of scalar intervention. We normalize the decision-maker's common agenda to $\mathbf{0}$. This is without loss because $A\mathbf{1} = \mathbf{1}$: any common target can be removed by recentering opinions. The matrix Q is symmetric and positive definite and weights opinion deviations. Sections 3 and 5 require

⁴The linear updating rule can also be interpreted locally. For a smooth rule $\mathbf{x}_{t+1} = F(\mathbf{x}_t) + \mathbf{b}r_t$, deviations around a baseline $\bar{\mathbf{x}}$ satisfy $\Delta \mathbf{x}_{t+1} \approx J(\bar{\mathbf{x}})\Delta \mathbf{x}_t + \mathbf{b}r_t$. The relevant local propagation matrix is therefore $J(\bar{\mathbf{x}})$; see Appendix S3.

⁵Because $(\mathbf{b}, r_t)$ and $(-\mathbf{b}, -r_t)$ generate the same intervention, the rule is identified only up to sign.

⁶The Euclidean bound is invariant to orthonormal changes of basis. Other institutional restrictions, such as nonnegativity, may generate corner solutions.

nothing further; Section 4 sets $Q = I$ so that losses do not mix eigendirections of A , and Section 6 takes Q diagonal with entries proportional to state population.

As a benchmark, under *full targeting* the decision-maker chooses an unrestricted intervention vector $\mathbf{u}_t \in \mathbb{R}^n$ in each period:

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{u}_t,$$

with intervention cost $c\|\mathbf{u}_t\|^2$. When $\|\mathbf{b}\| = 1$, a coarse intervention $\mathbf{u}_t = \mathbf{b}r_t$ has the same cost, $c\|\mathbf{u}_t\|^2 = cr_t^2$. The two technologies therefore differ only in the ability to change the exposure pattern over time. Let $v^f(\mathbf{x}_0)$ denote the full-targeting value. Section 5 characterizes the value of this targeting flexibility.

For a fixed exposure rule $\mathbf{b}$, let $v(\mathbf{x}_0; \mathbf{b})$ denote the value obtained when intervention levels are chosen optimally. The decision-maker chooses $\mathbf{b}$ to solve

$$\max_{\|\mathbf{b}\| \leq 1} v(\mathbf{x}_0; \mathbf{b}). \quad (1)$$

The exposure rule must therefore be chosen against the opinion path that it itself induces.

3 Characterizing optimal coarse targeting

Suppose opinions evolve only once, that is, $\mathbf{x}_1 = A\mathbf{x}_0 + \mathbf{b}r_0$, where $\|\mathbf{b}\| \leq 1$ and r_0 is a scalar message. The decision-maker chooses r_0 to solve

$$\max_{r_0} -\mathbf{x}_0' Q \mathbf{x}_0 - cr_0^2 - \delta \mathbf{x}_1' Q \mathbf{x}_1.$$

For a fixed exposure rule $\mathbf{b}$, the optimal message is linear:

$$r_0 = -\delta(c + \delta \mathbf{b}' Q \mathbf{b})^{-1} \mathbf{b}' Q A \mathbf{x}_0.$$

Since only the total intervention $\mathbf{u}_0 = \mathbf{b}r_0$ matters, the static problem is equivalent to choosing $\mathbf{u}_0$ directly. If $Q A \mathbf{x}_0 \neq \mathbf{0}$, the optimum satisfies $\|\mathbf{b}^*\| = 1$ and

$$\mathbf{b}^* r_0 = -\delta(cI + \delta Q)^{-1} Q A \mathbf{x}_0.$$

Thus fixing an exposure direction is costless in one period.

The restriction becomes substantive only when the same direction must be reused. For a fixed $\mathbf{b} \neq \mathbf{0}$, the decision-maker's inner problem is standard. The substantive problem is the outer choice of a rule that changes the path on which it will later be used. Its value is quadratic, so for a matrix $K^*(\mathbf{b})$ to be determined, write

$$v(\mathbf{x}; \mathbf{b}) = -\mathbf{x}'K^*(\mathbf{b})\mathbf{x}.$$

The Bellman equation is

$$v(\mathbf{x}; \mathbf{b}) = \max_r \left\{ -\mathbf{x}'Q\mathbf{x} - cr^2 - \delta(A\mathbf{x} + \mathbf{b}r)'K^*(\mathbf{b})(A\mathbf{x} + \mathbf{b}r) \right\}. \quad (2)$$

The optimal intervention level is linear,

$$r^*(\mathbf{x}; \mathbf{b}) = -\delta(c + \delta\mathbf{b}'K^*(\mathbf{b})\mathbf{b})^{-1}\mathbf{b}'K^*(\mathbf{b})A\mathbf{x} \equiv -\mathbf{p}(\mathbf{b})'\mathbf{x}, \quad (3)$$

and substitution gives the fixed-point equation

$$K^*(\mathbf{b}) = Q + \delta A' \left(K^*(\mathbf{b}) - \delta K^*(\mathbf{b})\mathbf{b}(c + \delta\mathbf{b}'K^*(\mathbf{b})\mathbf{b})^{-1}\mathbf{b}'K^*(\mathbf{b}) \right) A. \quad (4)$$

Lemma 1. *For every fixed $\mathbf{b} \neq \mathbf{0}$, $c > 0$, and $\delta \in (0, 1)$, there exists a unique symmetric positive definite matrix $K^*(\mathbf{b})$ solving (4). The optimal intervention level is given by (3), and the induced transition matrix $\tilde{A}(\mathbf{b})$ satisfies $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$.*

The induced transition matrix $\tilde{A}(\mathbf{b}) \equiv A - \mathbf{b}\mathbf{p}(\mathbf{b})'$ is stable and opinions evolve according to $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b})\mathbf{x}_t$. The exposure rule therefore plays two roles: it determines how the decision-maker intervenes, and it shapes the opinion path on which the same rule will later be used.

Lemma 2. *Suppose $QAx_0 \neq \mathbf{0}$. Then every optimal exposure rule satisfies $\|\mathbf{b}^*\| = 1$.*

Scaling up a fixed exposure direction implements the same intervention path with lower scalar-intervention cost. Since $QAx_0 \neq \mathbf{0}$ makes intervention strictly valuable, every optimum exhausts the capacity bound. The outer problem is therefore

$$\max_{\|\mathbf{b}\|=1} v(\mathbf{x}_0; \mathbf{b}) = -\mathbf{x}_0'K^*(\mathbf{b})\mathbf{x}_0, \quad K^*(\mathbf{b}) \text{ solves (4)}. \quad (5)$$

The object linking the two roles of the exposure rule is the discounted opinion-path matrix

$$Z(\mathbf{b}) = \sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b}))'^t. \quad (6)$$

It records where the path induced by $\mathbf{b}$ places its discounted weight: directions that are large or persistent receive greater weight. Standard perturbation results imply that $K^*(\mathbf{b})$, and hence $v(\mathbf{x}_0; \mathbf{b})$, vary smoothly with $\mathbf{b}$ and the primitives over the maintained parameter region (see Konstantinov et al. (1993) among others).

Proposition 1. *An optimal exposure rule exists. If $QAx_0 \neq 0$, any optimal exposure rule $\mathbf{b}^*$ satisfies, for some multiplier μ on $\|\mathbf{b}\| = 1$,*

$$\mu \mathbf{b}^* = K^*(\mathbf{b}^*) \tilde{A}(\mathbf{b}^*) Z(\mathbf{b}^*) \mathbf{p}(\mathbf{b}^*). \quad (7)$$

Proposition 1 captures the fixed point at the heart of coarse targeting.⁷ The exposure rule creates the path, and the path determines the marginal value of the rule. The matrix $Z(\mathbf{b}^*)$ summarizes where that rule will matter: directions that disappear quickly receive little weight, while directions carried forward by the network matter repeatedly. The remaining terms in (7) translate those directions into the value of reaching particular agents. Persistence determines what remains worth moving; leverage determines who can move it.

The optimal exposure direction is homogeneous in initial opinions: rescaling $\mathbf{x}_0$ by any nonzero scalar changes intervention intensity but not the optimal direction.

Because changing the exposure rule changes the path that gives the rule its value, the outer problem need not be concave and may have multiple stationary points. Under the nondegeneracy condition stated in Corollary S1 of the Supplementary Appendix, stationary points are finite and isolated, every local maximum is strict, and local stationary points vary smoothly with the primitives.

⁷The characterization extends to substochastic matrices A satisfying $\rho(A) < 1$. Row stochasticity is used in Section 4 to separate average opinion from disagreement.

4 Separating persistence from leverage

Persistent disagreement determines what remains worth moving; leverage determines where intervention is most effective. The two forces can be separated by shutting down one at a time: equal centrality removes differences in leverage, while immediate averaging removes persistent disagreement. We then combine them around a common benchmark.

Throughout this section, let $Q = I$. Equal weighting lets us separate persistence from leverage cleanly; with general payoff weights, the two forces remain present but need not separate. Denote the uniform vector, average opinion, and initial disagreement as

$$\mathbf{u} = \mathbf{1}/\sqrt{n}, \quad \bar{x}_0 = \mathbf{1}'\mathbf{x}_0/n, \quad \mathbf{d} = \mathbf{x}_0 - \bar{x}_0\mathbf{1} = (I - \mathbf{u}\mathbf{u}')\mathbf{x}_0.$$

4.1 Persistent disagreement under equal centrality

In a row-stochastic network, agents are equally central when $\mathbf{1}'A = \mathbf{1}'$, or equivalently, when A is doubly stochastic. This condition allows directed and asymmetric interactions. Because $A\mathbf{x}_0 = \bar{x}_0\mathbf{1} + A\mathbf{d}$, $A\mathbf{d}$ is precisely the disagreement that survives one round of updating.

Proposition 2. *Suppose A is doubly stochastic and $Q = I$, and suppose $\bar{x}_0 \neq 0$. Then uniform targeting is optimal if and only if $A\mathbf{d} = \mathbf{0}$.*

Under equal centrality, persistent disagreement is not merely one reason for nonuniform targeting; it is the only reason. Uniform exposure moves average opinion but not disagreement.⁸ If $A\mathbf{d} = \mathbf{0}$, inherited disagreement disappears immediately, so nonuniform exposure only creates new dispersion. If $A\mathbf{d} \neq \mathbf{0}$, some disagreement survives and a nonuniform rule can move it.

To obtain sharper comparative statics, the remainder of this subsection focuses on symmetric networks. Let

$$A = PDP', \quad D = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \mathbf{p}_1 = \mathbf{u},$$

and let $\tilde{\mathbf{x}} = P'\mathbf{x}$ and $\tilde{\mathbf{b}} = P'\mathbf{b}$. The first coordinate is average opinion; each $j > 1$ is a

⁸The assumption $\bar{x}_0 \neq 0$ ensures that there is an average-opinion component for uniform exposure to target.

disagreement dimension. The magnitude $|\lambda_j|$ measures how long that disagreement remains active, while its sign determines whether it maintains or reverses direction.⁹

Lemma 3. *Suppose A is symmetric and $Q = I$. If $\lambda_j \tilde{x}_0^j = 0$ for some $j > 1$, then any strict local optimum satisfies $\tilde{b}_j^* = 0$.*

Because $A\mathbf{d} = \sum_{j>1} \lambda_j \tilde{x}_0^j \mathbf{p}_j$, Proposition 2 determines whether any disagreement remains relevant, while Lemma 3 rules out exposure to dormant dimensions. A highly symmetric family makes this geometry transparent.

$$A(\rho) = \rho I + (1 - \rho) \frac{J}{n}, \quad \rho \in \left[-\frac{1}{n-1}, 1 \right]. \quad (8)$$

Here, $A(\rho)\mathbf{u} = \mathbf{u}$ and $A(\rho)\mathbf{d} = \rho\mathbf{d}$, so every disagreement dimension has the same persistence ρ .

Corollary 1. *Suppose $Q = I$, $A = A(\rho)$, and $\mathbf{d} \neq \mathbf{0}$. Then any strict local maximum satisfies $\mathbf{b}^*(\rho) \in \text{span}\{\mathbf{u}, \mathbf{d}\}$.*

Let $\mathbf{b}^*(\rho) = \alpha\mathbf{u} + \beta \frac{\mathbf{d}}{\|\mathbf{d}\|}$. The induced dynamics reduce to

$$m_{t+1} = m_t + \alpha r_t, \quad y_{t+1} = \rho y_t + \beta r_t,$$

where $m_t = \mathbf{u}'\mathbf{x}_t$ and y_t is disagreement along $\mathbf{d}$. The fixed rule therefore divides its capacity between average opinion and $\mathbf{d}$, the single initially active direction of disagreement. At $\rho = 0$, disagreement disappears after one update and optimal exposure is uniform. At $\rho = 1$, no disagreement is removed, so optimal exposure is proportional to $\mathbf{x}_0$.

The two-island example in the introduction has the same structure. With n agents in each island, weight p_s on each same-island opinion, and weight p_d on each cross-island opinion, the group-divide dimension has eigenvalue $n(p_s - p_d)$, while every within-island dimension has eigenvalue zero. By Lemma 3, any strict local optimum ignores within-island disagreement, regardless of its magnitude, and assigns identical exposure within each island.

⁹The change of basis rotates the state and exposure vector, not the instrument: the decision-maker still chooses one scalar intervention level r_t each period.

4.2 Agent leverage without persistent disagreement

We now isolate leverage by eliminating inherited disagreement. In a uniform opinion-leader network, absent intervention, all opinions collapse after one update. Unequal exposure therefore reflects differences in agents' leverage over future opinions.

Definition 1. A network is a **uniform opinion-leader network** if $A = \mathbf{1}\mathbf{a}'$, where $\mathbf{a}'\mathbf{1} = 1$ and $a_1 > a_2 > \dots > a_n > 0$.

Every agent listens to the same authority-weighted average $\mathbf{a}'\mathbf{x}$. Without intervention, $A\mathbf{x} = (\mathbf{a}'\mathbf{x})\mathbf{1}$, so the network carries no inherited disagreement beyond one update.

Proposition 3. Fix $\delta \in (0, 1)$ and $c > 0$, and suppose $Q = I$. For every initial opinion satisfying $\mathbf{a}'\mathbf{x}_0 \neq 0$, choose the sign normalization $\mathbf{a}'\mathbf{b}^* > 0$. The optimal exposure direction is independent of $\mathbf{x}_0$ and has the form

$$\mathbf{b}^* = \eta\mathbf{1} + \kappa\mathbf{a}, \quad \eta > 0, \quad \kappa > 0.$$

Consequently,

$$b_1^* > \dots > b_n^* > 0, \quad b_i^* - b_j^* = \kappa(a_i - a_j).$$

Exposure to high-authority agents moves the future consensus more, pulling the rule toward $\mathbf{a}$, while concentrated exposure creates temporary dispersion, pulling it toward $\mathbf{1}$. The optimum therefore follows authority without concentrating on the leader: every agent remains active, $b_i^* > 0$, even when one agent commands almost all authority.

4.3 Persistence and leverage together

We now combine persistence and leverage around a common benchmark. Let $A_0 = PDP'$ be symmetric and row-stochastic, with $\mathbf{p}_1 = \mathbf{u}$. Using the homogeneity of the optimal exposure direction, normalize the benchmark average-opinion component to one and perturb opinions and the network according to

$$\tilde{\mathbf{x}}_0 = \mathbf{e}_1 + \boldsymbol{\xi}, \quad A_\varepsilon = A_0 + \varepsilon B, \quad B\mathbf{1} = \mathbf{0},$$

restricting attention to sufficiently small $\boldsymbol{\xi}$ and $|\varepsilon|$ such that A_ε remains nonnegative.

Define $\mathbf{s} = B'\mathbf{u}$. The component s_i is the first-order change in the aggregate weight placed on agent i 's opinion: $s_i > 0$ means that the perturbation increases this weight, while $s_i < 0$ means that it decreases it. Moreover, $\mathbf{u}'\mathbf{s} = 0$, so the perturbation redistributes influence across agents without changing its total.

At the benchmark, uniform exposure is optimal. Let $\tau \in (0, 1)$ denote the induced persistence of the average-opinion coordinate under the optimal intensity policy, so that the average opinion evolves as $\tilde{x}_{t+1}^1 = \tau \tilde{x}_t^1$.

Proposition 4. *Fix (δ, c) . Suppose $Q = I$ and $\boldsymbol{\xi} \perp \mathbf{e}_1$, and define*

$$\mathbf{g} = (I - \delta\tau A'_0)^{-1}\mathbf{s} = \sum_{j>1} g_j \mathbf{p}_j.$$

There exist functions $M : [-1, 1] \rightarrow \mathbb{R}$ and $\chi : [-1, 1] \rightarrow \mathbb{R}_{++}$, with M strictly increasing, such that the unique strict local optimum near $\mathbf{e}_1$ satisfies

$$\tilde{\mathbf{b}}^*(\boldsymbol{\xi}, \varepsilon) = \mathbf{e}_1 + \sum_{j>1} [M(\lambda_j)\xi_j + \varepsilon\chi(\lambda_j)g_j] \mathbf{e}_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Moreover, $M(0) = 0$, $M(1) = 1$, and $0 < M(\lambda) < \lambda$ for all $\lambda \in (0, 1)$.

Setting $\varepsilon = 0$ isolates persistence. In the one-period problem, dimension j receives exposure coefficient $\lambda_j \xi_j$:

$$\tilde{\mathbf{b}}^{\text{static}}(\boldsymbol{\xi}) = \mathbf{e}_1 + \sum_{j>1} \lambda_j \xi_j \mathbf{e}_j + o(\|\boldsymbol{\xi}\|).$$

Under reuse, the corresponding coefficient is $M(\lambda_j)\xi_j$. A reusable rule therefore tilts less toward positively persistent disagreement, because the same exposure pattern remains in place after that disagreement has decayed, crowding out exposure to the enduring average-opinion dimension. The wedge is strict for $\lambda \in (0, 1)$ and disappears at the endpoints: when $\lambda = 0$ the disagreement is immediately irrelevant, while at $\lambda = 1$ it is as persistent as average opinion.¹⁰ For $\lambda < 0$, the surviving disagreement reverses sign, so exposure tilts against its current direction.

The two-island example in the introduction illustrates the same force away from the local benchmark: its between-island disagreement has eigenvalue $\lambda = 0.6$, and reuse correspond-

¹⁰The reuse wedge vanishes as intervention becomes cheap: as $c \downarrow 0$, $M(\lambda) \rightarrow \lambda$. As $c \rightarrow \infty$, $M(\lambda) \rightarrow \lambda(1 - \delta)^2 / [(1 - \delta\lambda^2)(1 - \delta\lambda)]$.

ingly dampens the tilt toward that dimension.

Setting $\xi = \mathbf{0}$ isolates leverage. The decision-maker responds not to the direct redistribution $\mathbf{s}$, but to its propagated value

$$\mathbf{g} = \mathbf{s} + \delta\tau A'_0 \mathbf{s} + \delta^2 \tau^2 (A'_0)^2 \mathbf{s} + \dots$$

Thus $\mathbf{g}$ is a signed Katz–Bonacich measure of effective leverage, with $\delta\tau$ reflecting discounting and the decline of the average-opinion benchmark under optimal intervention. Persistence and leverage add to first order; their interaction enters only at higher order.

The clique connects Proposition 4 to Proposition 3. Let

$$A_\varepsilon = \mathbf{1}\mathbf{a}'_\varepsilon, \quad \mathbf{a}_\varepsilon = (1 - \varepsilon)\frac{1}{n}\mathbf{1} + \varepsilon\mathbf{a},$$

so that $A_0 = J/n$ and

$$\mathbf{s} = \sqrt{n} \left(\mathbf{a} - \frac{1}{n}\mathbf{1} \right).$$

Since $A'_0 \mathbf{s} = \mathbf{0}$ and every disagreement eigenvalue of A_0 is zero, $\mathbf{g} = \mathbf{s}$ and $\chi(\lambda_j) = \chi(0) > 0$. Hence

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon C \left(\mathbf{a} - \frac{1}{n}\mathbf{1} \right) + o(|\varepsilon|), \quad C > 0,$$

recovering locally the authority ordering in Proposition 3.

5 The value of targeting flexibility

The same endogenous path that determines the optimal coarse rule also determines the value of relaxing it. Recall the full-targeting benchmark from Section 2, under which the decision-maker can choose an unrestricted intervention vector each period. Full flexibility is valuable only when the path induced by a coarse rule visits directions in which one fixed exposure rule cannot reproduce the full intervention.

Let $v^f(\mathbf{x}_0)$ denote the full-targeting value. Since every coarse-targeting path is feasible under full targeting, $v^f(\mathbf{x}_0) \geq v(\mathbf{x}_0; \mathbf{b})$. Define

$$\Delta(\mathbf{b}) = v^f(\mathbf{x}_0) - v(\mathbf{x}_0; \mathbf{b}) \geq 0, \quad \Delta^* = \min_{\|\mathbf{b}\|=1} \Delta(\mathbf{b}).$$

To characterize the gap, compare the intervention matrices. Write the full-targeting value as

$$v^f(\mathbf{x}) = -\mathbf{x}' K^f \mathbf{x}.$$

Under full targeting,

$$\mathbf{u}_t = -F^f \mathbf{x}_t, \quad F^f = \delta(cI + \delta K^f)^{-1} K^f A.$$

Under coarse targeting,

$$\mathbf{u}_t = -F^c(\mathbf{b}) \mathbf{x}_t, \quad F^c(\mathbf{b}) = \mathbf{b} \mathbf{p}(\mathbf{b})'.$$

Define

$$\Phi(\mathbf{b}) = (F^c(\mathbf{b}) - F^f)'(cI + \delta K^f)(F^c(\mathbf{b}) - F^f).$$

The matrix $\Phi(\mathbf{b})$ measures the cost of intervention error, while $Z(\mathbf{b})$ records where the coarse path places discounted weight.

Proposition 5. *For any exposure rule satisfying $\|\mathbf{b}\| = 1$,*

$$\Delta(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_t' \Phi(\mathbf{b}) \mathbf{x}_t = \text{tr}(\Phi(\mathbf{b}) Z(\mathbf{b})),$$

where $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b}) \mathbf{x}_t$, $t \geq 0$.

A fixed one-dimensional rule can attain the full-targeting value even when its policy differs from the full policy away from the realized path. Since $cI + \delta K^f$ is positive definite,

$$\Delta(\mathbf{b}) = 0 \iff (F^c(\mathbf{b}) - F^f) \mathbf{z} = \mathbf{0} \quad \text{for every } \mathbf{z} \in \text{Range}(Z(\mathbf{b})).$$

Thus exact equivalence requires matching full targeting only on the path induced by the coarse rule itself. Because $Z(\mathbf{b})$ depends on $\mathbf{b}$, this is an endogenous fixed-point condition.

To measure approximate performance, write

$$Z(\mathbf{b}) = \sum_{k=1}^n \sigma_k \mathbf{q}_k \mathbf{q}_k', \quad \sigma_1 \geq \dots \geq \sigma_n \geq 0.$$

Then

$$\Delta(\mathbf{b}) = \sum_{k=1}^n \sigma_k \mathbf{q}'_k \Phi(\mathbf{b}) \mathbf{q}_k.$$

Thus the gap is small when discounted path weight is concentrated in a few directions and the coarse and full policies agree closely on those directions. A convenient scalar summary of path concentration is the effective path dimension

$$d_{\text{eff}}(Z(\mathbf{b})) = \frac{\text{tr}(Z(\mathbf{b}))^2}{\text{tr}(Z(\mathbf{b}))}. \quad (9)$$

It equals one for a rank-one path and increases as discounted path weight spreads across directions.¹¹

Exact implementation asks how many reusable directions are needed to replicate full targeting. Suppose intervention is restricted to a fixed subspace $\mathcal{S}$, and let $\{\mathbf{x}_t^f\}$ denote the opinion path under full targeting from $\mathbf{x}_0$.

Corollary 2. *A fixed subspace $\mathcal{S}$ attains the full-targeting value if and only if $F^f \mathbf{x}_t^f \in \mathcal{S}$ for every $t \geq 0$. Hence the minimum dimension required to implement full targeting is*

$$k_{\min} = \text{rank}(F^f Z^f F^{f'}) , \quad Z^f = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_t^f \mathbf{x}_t^{f'}.$$

Because $\text{Range}(Z^f) = \text{span}\{\mathbf{x}_t^f : t \geq 0\}$, $k_{\min}$ is precisely the dimension of $\text{span}\{F^f \mathbf{x}_t^f : t \geq 0\}$. Exact implementation may require many directions even when one captures almost all attainable gains.

A complementary primitive implication is that retargeting cannot be very valuable when the network quickly eliminates disagreement.

Corollary 3. *Suppose A is symmetric and $Q = I$. Let $\mathbf{d}$ denote initial disagreement and let $|\lambda_2|$ be the largest absolute eigenvalue of A on the disagreement space. Then*

$$0 \leq \Delta^* \leq \frac{\delta |\lambda_2|^2}{1 - \delta |\lambda_2|^2} \|\mathbf{d}\|^2.$$

¹¹This differs from the one-dimensionality in Galeotti et al. (2020), which comes from concentration on a principal component of a static network game; here low dimensionality is a property of the endogenous path induced by the intervention rule.

To see the bound, consider the feasible uniform rule. Under symmetry and $Q = I$, it solves the same scalar problem as full targeting in the average-opinion dimension, while disagreement evolves as $A^t \mathbf{d}$. Since the date-zero loss is common, the value gap is at most the subsequent loss from uncontrolled disagreement:

$$\Delta^* \leq \Delta(\mathbf{u}) \leq \sum_{t=1}^{\infty} \delta^t \|A^t \mathbf{d}\|^2 \leq \frac{\delta |\lambda_2|^2}{1 - \delta |\lambda_2|^2} \|\mathbf{d}\|^2.$$

Thus fast elimination of disagreement can make flexibility inexpensive even when the realized path is not rank one; effective path dimension also depends on initial opinions and the induced policy. Section 6 evaluates the two diagnostics from the trace identity: path concentration and the gap generated along its leading directions.

6 Empirical illustration: climate-opinion disagreement

We illustrate the theory using U.S. state-level climate opinions and the Facebook Social Connectedness Index (SCI). We do not estimate opinion dynamics; we use observed disagreement and social connectedness to compute the exposure rule and the value of targeting flexibility implied by the model.

6.1 Data and construction

We construct $x_{0,s}$ by population-weighting county `consensusOppose` estimates from the 2023 Yale Climate Opinion Maps (Howe et al., 2015; Marlon et al., 2022). It is the estimated share of residents in state s who disagree with the statement “Most scientists think global warming is happening.” We interpret this as disagreement with the scientific-consensus benchmark and normalize the decision-maker’s agenda to zero. Figure 1 shows the resulting opinion vector.

We measure network links using the Facebook SCI, aggregated from counties to states following Bailey et al. (2021). The construction derived in Appendix S5 gives

$$\hat{A}_{ss'} = \frac{\text{SCI}_{ss'} N_{s'}}{\sum_{s''} \text{SCI}_{ss''} N_{s''}},$$

where $N_{s'}$ is the population of destination state s' . Thus $\hat{A}_{ss'}$ is the weight state s places on

Percentage of people who disagree with
"Most scientists think global warming is happening"

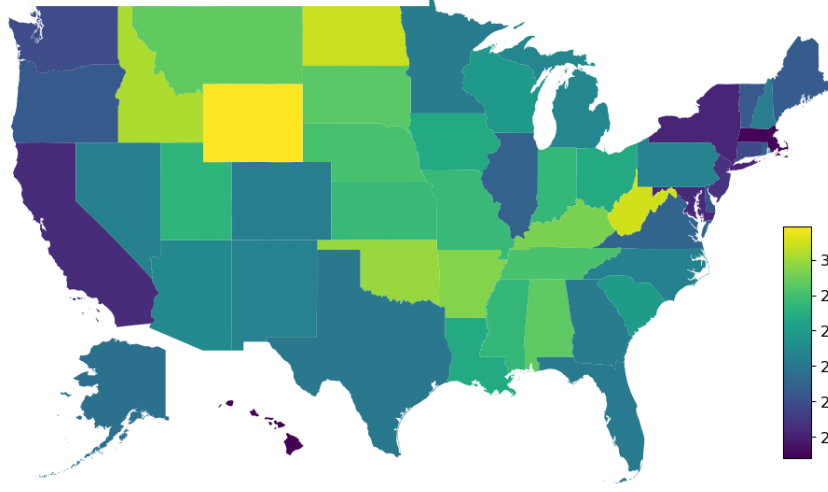


Figure 1: Disagreement with the scientific consensus across U.S. states.

opinions in state s' . The resulting row-stochastic network is asymmetric and geographically clustered, as shown in Figure 2. Authority and persistent disagreement therefore need not point toward the same states: a state may have high leverage over future opinions, or it may be connected to disagreement that the network continues to carry forward.

6.2 Which states receive exposure?

We compute the highest-value stationary coarse rule found by the multi-start procedure in Appendix S5.2, denoted $\hat{\mathbf{b}}$. Our baseline uses $\delta = 0.9$, $c = 10$, and population-weighted losses,

$$Q^{\text{pop}} = \text{diag}(N_s / \bar{N}),$$

where $\bar{N}$ is average state population. Population enters both through the construction of $\hat{A}$ and through Q^{pop} , which weights each state's loss by the residents it represents.

We use the leading right singular vector of $\hat{A}$, denoted by $\mathbf{v}_1$ and normalized to unit Euclidean norm, as the authority benchmark.¹² As before, let $\mathbf{d} = \mathbf{x}_0 - \bar{x}_0 \mathbf{1}$ denote centered cross-state

¹²The state-ordering is unchanged if authority is instead measured by the invariant distribution or column sums of $\hat{A}$. California remains above Texas, and Ohio remains below both.

0.398 to 0.325.¹³ Thus reuse materially reallocates exposure even though, as the next subsection shows, its value gain is small. Table 1 summarizes the baseline comparisons.

Table 1: Authority, future persistence, and exposure at the baseline ($\delta = 0.9$, $c = 10$). All three are normalized to unit Euclidean norm; future persistence is signed.

	Texas	California	Ohio
Authority	0.3430	0.4140	0.1600
Future persistence	-0.0760	-0.1500	-0.0230
Reusable exposure	0.3248	0.3280	0.2195
One-period exposure	0.3983	0.4366	0.2096
Reusable minus one-period	-0.0735	-0.1086	0.0099

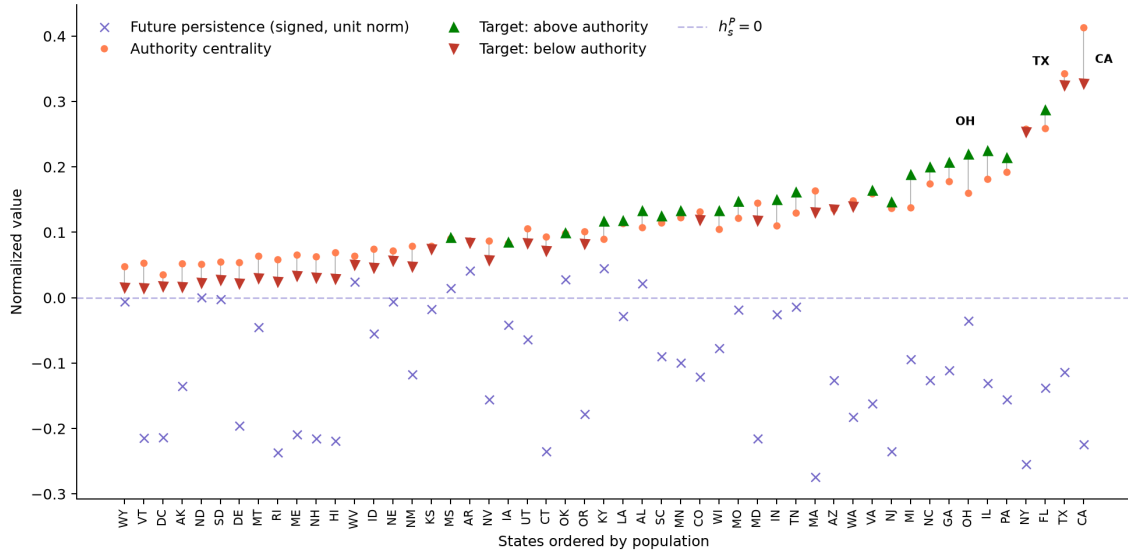


Figure 3: Population-weighted exposure compared with authority and future persistence across states. Authority and exposure have unit Euclidean norm; future persistence is signed and normalized to unit Euclidean norm. States are ordered by population, and California, Texas, and Ohio are labeled.

6.3 How valuable is targeting flexibility?

We compare the reported coarse rule with full targeting, fixed exposure directions, and period-by-period myopic retargeting.¹⁴ Fixed directions use their optimal dynamic scalar

¹³Appendix Table 5 reports the largest state-level reallocations, including increases for Mississippi, Minnesota, and Alabama.

¹⁴Because $\hat{\mathbf{b}}$ is feasible, its reported capture is a lower bound on optimal coarse capture.

policies, while the myopic policy reoptimizes an unrestricted one-period vector intervention each period. For a rule with value v , define its share of attainable gains as

$$\frac{v - v^\varnothing}{v^f - v^\varnothing}.$$

Table 2 shows that across the displayed intervention costs, the reported coarse rule captures

Table 2: Share of attainable gains captured ($\delta = 0.9$, $Q = Q^{\text{pop}}$). Entries are rounded to the nearest tenth of a percentage point.

	$c = 1$	$c = 10$	$c = 100$
Opinion ($\mathbf{b} \propto \mathbf{x}_0$)	98.3%	91.1%	72.0%
Authority ($\mathbf{b} \propto \mathbf{v}_1$)	97.2%	99.1%	97.8%
One-period direction	99.6%	99.2%	99.5%
Period-by-period myopic retargeting	99.7%	91.5%	52.2%
Coarse ($\mathbf{b} = \hat{\mathbf{b}}$)	99.9%	99.9%	100.0%

at least 99.9% of attainable gains, while period-by-period myopic retargeting falls to 52.2% by $c = 100$. Flexibility does not compensate for ignoring continuation values. The one-period direction also remains close to the reusable rule in value despite the substantial reallocation under reuse documented in Section 6.2. This is consistent with stationarity of value at the optimal reusable direction: nearby changes in direction have only second-order welfare effects. The Supplementary Appendix reports the wider cost grid and additional fixed heuristics.

Proposition 5 gives the mechanism:

$$\Delta(\hat{\mathbf{b}}) = \text{tr}\left(\Phi(\hat{\mathbf{b}})Z(\hat{\mathbf{b}})\right).$$

At the baseline, the realized path has effective dimension 1.03. Although its leading direction carries 98.59% of discounted path weight, that direction contributes only 1.14% of the coarse-to-full gap. Thus the policies disagree particularly little on the direction that matters most.

The common component accounts for 98.0% of $\|\mathbf{x}_0\|^2$. To test whether it drives path concentration, we replace $\mathbf{x}_0$ by $\mathbf{d}$ and recompute the coarse and full policies, holding $\hat{A}$, Q^{pop} , δ , and c fixed. Concentration survives centering. Starting from observed cross-state

Table 3: Baseline and centered-disagreement paths ($\delta = 0.9$, $c = 10$, $Q = Q^{\text{pop}}$).

Initial opinion	Coarse capture	$d_{\text{eff}}(Z)$	Lead share
$\mathbf{x}_0$	99.91%	1.03	98.59%
$\mathbf{d}$	99.61%	1.08	96.24%

disagreement, the feasible coarse rule still captures 99.61% of attainable gains. Thus the low value of retargeting is not confined to initial opinions close to the average-opinion direction.

The concentration is not mechanical: in centered random row-stochastic networks with median second-eigenvalue modulus 0.77, close to Facebook’s 0.76, median effective dimension is 2.25 and the median leading share is 62%, far less concentrated than the centered Facebook path in Table 3.

Recurrent shocks provide the boundary. The cross-state standard deviation of initial disagreement is about 0.036, so $\sigma = 0.1$ is already almost three times the empirical dispersion. Coarse targeting still captures 97.2% of attainable gains at that shock size, but loses value as shocks populate new directions; Appendix Table 7 reports the full comparison. Retargeting becomes valuable when disagreement is diffuse rather than concentrated and predictable.

7 The dimension of new disagreement

We now ask what makes path weight diffuse. Recurrent shocks inject disagreement in new directions, while a persistent bias repeatedly injects the same source. In the highly symmetric benchmark, the latter adds at most one relevant disagreement direction.

7.1 Recurrent shocks and dimensional flexibility

Suppose recurrent agent-level shocks are iid and arrive after intervention:

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{u}_t + \boldsymbol{\nu}_t, \quad \boldsymbol{\nu}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I).$$

The decision-maker observes $\mathbf{x}_t$ before choosing $\mathbf{u}_t$. We restrict attention to symmetric networks and write

$$A = PDP', \quad \tilde{\mathbf{x}}_t = P'\mathbf{x}_t, \quad \tilde{\boldsymbol{\nu}}_t = P'\boldsymbol{\nu}_t.$$

Since P is orthonormal, $\tilde{\nu}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$.

We compare three rules under the same intervention-cost normalization. Under *full targeting*, the decision-maker chooses an unrestricted vector $\tilde{\mathbf{u}}_t$. Under *coarse targeting*, intervention is restricted to $\tilde{\mathbf{u}}_t = \tilde{\mathbf{b}}r_t$, with $\|\tilde{\mathbf{b}}\| = 1$ fixed at $t = 0$. Under *myopic targeting*, the decision-maker retains the unrestricted vector technology but treats the next period as terminal:

$$\tilde{\mathbf{u}}_t^m \in \arg \min_{\tilde{\mathbf{u}}} \{c\|\tilde{\mathbf{u}}\|^2 + \delta\|D\tilde{\mathbf{x}}_t + \tilde{\mathbf{u}}\|^2\}.$$

Thus

$$\tilde{\mathbf{u}}_t^m = -\frac{\delta}{c + \delta} D\tilde{\mathbf{x}}_t.$$

Myopic targeting retains dimensional flexibility but ignores the persistence of the disagreement it moves.

Because the shocks are additive and mean zero, they do not change the optimal response to current opinions. For each rule $q \in \{f, m, c\}$,

$$\tilde{v}^q(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^q \tilde{\mathbf{x}}_0 - \frac{\delta \sigma^2}{1 - \delta} \text{tr}(\tilde{K}^q),$$

where the two terms capture losses from initial opinions and recurrent shocks, respectively.

Proposition 6. *Suppose A is symmetric, $Q = I$, $\nu_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$, and $\sigma > 0$.*

- (1) *Full targeting strictly dominates myopic targeting.*
- (2) *Full targeting weakly dominates coarse targeting, with equality if $\text{rank}(A) = 1$ and strict dominance if $\text{rank}(A) > 1$.*

The comparison separates foresight from dimensional flexibility. The substantive question is which advantage matters more. We study the tradeoff in the highly symmetric family in (8).

Proposition 7. *Suppose $Q = I$, $A = A(\rho)$, $\sigma > 0$ and $\mathbf{d} \neq \mathbf{0}$.*

- (1) *For each fixed n , an optimal coarse exposure rule can be chosen in $\text{span}\{\mathbf{u}, \mathbf{d}\}$.*
- (2) *Fix $\rho \in (0, 1]$ and consider a sequence of such networks as $n \rightarrow \infty$. If $|\mathbf{u}'_n \mathbf{x}_0^{(n)}| = O(1)$ and $\|\mathbf{d}_n\| = O(1)$, then*

$$1 > \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^m} > \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^c} > 0.$$

Part (1) shows that the span restriction from Section 4.1 survives recurrent shocks. The iid shocks make all disagreement directions orthogonal to $\mathbf{d}$ ex ante symmetric, so an optimal fixed rule need not select among them. The shocks nevertheless realize disagreement in every one of those directions.

Part (2) delivers a reversal: as the network grows, myopic targeting comes closer to full targeting than coarse targeting does. Recurrent shocks populate every disagreement direction, so losses in directions omitted by a fixed rule eventually dominate despite its foresight.

The richer fixed-subspace technology from Section 5 gives the same boundary quantitatively. If its dimension is k_n and $k_n/n \rightarrow \alpha$, then it captures asymptotically an α share of the gains from controlling recurrent disagreement versus no intervention. Thus preserving a fixed share of the gains requires the number of reusable directions to grow proportionally with the number of disagreement dimensions.

The polar cases make the mechanism transparent. In the clique ($\rho = 0$), disagreement disappears after one update and coarse targeting matches full targeting. In the identity network, the deterministic path is concentrated, but recurrent shocks populate every disagreement direction, so a fixed rule loses value relative to full targeting as σ increases (Figure 4).

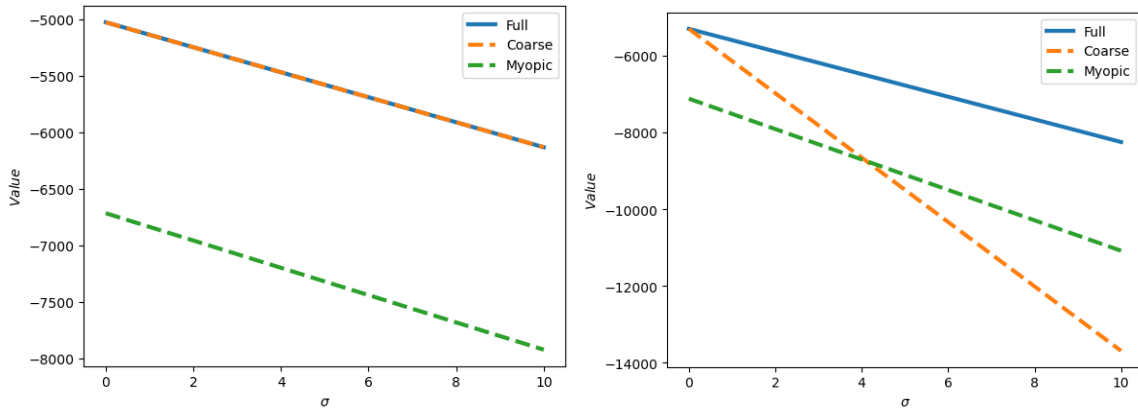


Figure 4: Value comparison across targeting methods in the clique (left) and the identity network (right) as the scale of recurrent shocks increases.

7.2 Persistent bias and predictable disagreement

Now consider a time-invariant bias,¹⁵

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t + \mathbf{w}.$$

For any fixed exposure rule, the optimal intervention is affine in current opinions. Repeated injection of the same source generates the predictable network orbit $\mathbf{w}, A\mathbf{w}, A^2\mathbf{w}, \dots$. Proposition S1 in the appendix shows that optimal exposure need only respond to this orbit and to inherited disagreement that survives social updating. In the highly symmetric family, the dimensional implication is sharp.

Corollary 4. *Suppose $Q = I$ and $A = A(\rho)$. Let $\mathbf{d} = (I - \mathbf{u}\mathbf{u}')\mathbf{x}_0$, $\mathbf{w}_\perp = (I - \mathbf{u}\mathbf{u}')\mathbf{w}$. Then every strict local optimum satisfies*

$$\mathbf{b}^* \in \text{span}\{\mathbf{u}, \rho\mathbf{d}, \mathbf{w}_\perp\}.$$

Under $A(\rho)$, inherited disagreement contributes $\rho\mathbf{d}$, while the persistent bias contributes $\mathbf{w}_\perp$. Thus a persistent source adds at most one relevant disagreement direction.

The clique gives the sharpest example. When $\rho = 0$, inherited disagreement disappears after one update, but $\mathbf{w}_\perp$ is replenished every period. Hence $\mathbf{b}^* \in \text{span}\{\mathbf{u}, \mathbf{w}_\perp\}$, so a persistent source can create nonuniform targeting without creating high-dimensional disagreement.

8 Conclusion

When an exposure rule must be reused, the relevant object is not a ranking of agents but the geometry of the opinion path. Optimal exposure follows the disagreement the network preserves and the agents who have leverage over it.

More broadly, the cost of a simple instrument is governed not by the size of the system, but by the directions that matter along the path on which the instrument will be used. Reuse is inexpensive when those directions are few and predictable; flexibility becomes valuable when new disagreement is diffuse. The same logic extends to intermediate technologies with several slowly adjustable exposure rules.

¹⁵The vector $\mathbf{w}$ can represent persistent misinformation, agent-level bias, or anchored opinions.

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A Proofs

Standard or subsidiary results and proofs are relegated to Section S1 of the Supplementary Appendix. They are Lemma 1, Lemma 2, Corollary S1, Lemma 3, Corollary 1 and Lemma 4.

A.1 Proofs from Section 3

Proof of Proposition 1: We derive the gradient of the value with respect to the exposure rule. Fix $\mathbf{b} \neq \mathbf{0}$. By Lemma 1, the optimal intervention level is unique and linear: there exists a vector $\mathbf{p}$ such that $r_t = -\mathbf{p}'\mathbf{x}_t$ for all $t \geq 0$. Under this policy, $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b})\mathbf{x}_t$ where $\tilde{A}(\mathbf{b}) \equiv A - \mathbf{b}\mathbf{p}'$. By Lemma 1, $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$, so all discounted sums below are well defined. For any opinion $\mathbf{x}_t$, the Bellman equation evaluated at the optimal r_t is

$$v(\mathbf{x}_t; \mathbf{b}) = -\mathbf{x}_t' Q \mathbf{x}_t - c r_t^2 - \delta \mathbf{x}_{t+1}' K^*(\mathbf{b}) \mathbf{x}_{t+1}, \text{ and } \mathbf{x}_{t+1} = A \mathbf{x}_t + \mathbf{b} r_t.$$

By the Envelope theorem, terms involving $\partial r_t / \partial \mathbf{b}$ vanish. Holding the current opinion fixed, $\partial \mathbf{x}_{t+1} / \partial \mathbf{b} = r_t I$. Differentiating with respect to $\mathbf{b}$,

$$\nabla_{\mathbf{b}} v(\mathbf{x}_t; \mathbf{b}) = -2\delta r_t K^*(\mathbf{b}) \mathbf{x}_{t+1} + \delta \nabla_{\mathbf{b}} v(\mathbf{x}_{t+1}; \mathbf{b}).$$

Iterating forward from $t = 0$ and using $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$ gives

$$\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = -2\delta \sum_{t=0}^{\infty} \delta^t r_t K^*(\mathbf{b}) \mathbf{x}_{t+1}.$$

Substituting $r_t = -\mathbf{p}' \tilde{A}(\mathbf{b})^t \mathbf{x}_0$ and $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b})^{t+1} \mathbf{x}_0$ yields

$$\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = 2\delta K^*(\mathbf{b}) \tilde{A}(\mathbf{b}) \left(\sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b})')^t \right) \mathbf{p}.$$

Recall that $Z \equiv \sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b})')^t$. Then $\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = 2\delta K^*(\mathbf{b}) \tilde{A}(\mathbf{b}) Z \mathbf{p}$. Moreover, because $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$, Z is the unique solution to the Lyapunov equation

$$Z = \delta \tilde{A}(\mathbf{b}) Z \tilde{A}(\mathbf{b})' + \mathbf{x}_0 \mathbf{x}_0'.$$

The Lagrangian for the constrained problem on the unit sphere is

$$\mathcal{L}(\mathbf{b}, \lambda) = v(\mathbf{x}_0; \mathbf{b}) - \lambda(\mathbf{b}'\mathbf{b} - 1).$$

At a constrained stationary point on the sphere, $\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = 2\lambda \mathbf{b}$. Using the gradient expression above and absorbing the positive scalar 2δ into the multiplier gives $\mu \mathbf{b} = K^*(\mathbf{b}) \tilde{A}(\mathbf{b}) Z \mathbf{p}$, which is (7). At $\mathbf{b} = \mathbf{0}$, define

$$K^*(\mathbf{0}) = \sum_{t=0}^{\infty} \delta^t (A')^t Q A^t,$$

the no-intervention loss matrix. The fixed-rule value extends continuously to $\mathbf{b} = \mathbf{0}$. Hence $v(\mathbf{x}_0; \mathbf{b})$ is continuous on the compact feasible set $\{\mathbf{b} : \|\mathbf{b}\| \leq 1\}$. Existence follows from continuity of $v(\mathbf{x}_0; \mathbf{b})$ and compactness of the feasible set $\{\mathbf{b} : \|\mathbf{b}\| \leq 1\}$. If $Q A \mathbf{x}_0 \neq \mathbf{0}$, Lemma 2 implies that every optimizer satisfies $\|\mathbf{b}^*\| = 1$. Hence every optimizer satisfies the first-order condition above. $\square$

A.2 Proofs from Section 4

Proof of Proposition 2: Let $\Pi = I - \mathbf{u}\mathbf{u}'$ denote the projection onto deviations from average opinion. Decompose any exposure rule as $\mathbf{b} = \alpha \mathbf{u} + \mathbf{q}$, $\alpha = \mathbf{u}'\mathbf{b}$, and $\mathbf{q} \perp \mathbf{u}$.

Recall $\bar{x}_t = \frac{1}{n} \mathbf{1}' \mathbf{x}_t = \frac{1}{\sqrt{n}} \mathbf{u}' \mathbf{x}_t$. Because A is doubly stochastic, $A\mathbf{u} = \mathbf{u}$ and $\mathbf{u}'A = \mathbf{u}'$. Therefore, under $(\mathbf{b}, \{r_t\})$, average opinion evolves according to

$$\bar{x}_{t+1} = \bar{x}_t + \frac{\alpha}{\sqrt{n}} r_t.$$

Suppose first that $A\mathbf{d} = \mathbf{0}$. Given any feasible exposure rule $\mathbf{b} = \alpha\mathbf{u} + \mathbf{q}$ and intervention sequence $\{r_t\}$, consider uniform exposure $\mathbf{u}$ with $\hat{r}_t = \alpha r_t$. Thus $\bar{x}_{t+1} = \bar{x}_t + \frac{1}{\sqrt{n}} \hat{r}_t$. The two policies generate the same path of average opinion. Moreover, $A\mathbf{d} = \mathbf{0}$, so inherited disagreement disappears after the first update. Uniform exposure creates no new disagreement, whereas a nonuniform exposure rule may do so. Since

$$\|\mathbf{x}_t\|^2 = n\bar{x}_t^2 + \|\Pi\mathbf{x}_t\|^2,$$

uniform exposure therefore generates weakly lower opinion loss at every date. Its intervention cost is also weakly lower because $\hat{r}_t^2 = \alpha^2 r_t^2 \leq r_t^2$, where $|\alpha| \leq \|\mathbf{b}\| \leq 1$. Thus uniform targeting weakly dominates every exposure rule and is optimal.

Conversely, suppose that $A\mathbf{d} \neq \mathbf{0}$ and consider uniform exposure, $\mathbf{b} = \mathbf{u}$. Since both $\text{span}\{\mathbf{u}\}$ and $\mathbf{u}^\perp$ are invariant under A and A' , the Riccati map preserves matrices that do not mix the average-opinion direction and the disagreement space. By uniqueness of the stabilizing Riccati solution,

$$K^* = \tilde{k}_1 \mathbf{u}\mathbf{u}' + K_\perp, \quad K_\perp = \Pi K^* \Pi,$$

where $K_\perp$ is positive definite on $\mathbf{u}^\perp$. Moreover,

$$\mathbf{p} = p_1 \mathbf{u}, \quad \tilde{A}(\mathbf{u}) = a_1 \mathbf{u}\mathbf{u}' + A_\perp, \quad A_\perp = \Pi A \Pi,$$

where

$$p_1 = \frac{\delta \tilde{k}_1}{c + \delta \tilde{k}_1} > 0, \quad a_1 = 1 - p_1 = \frac{c}{c + \delta \tilde{k}_1} \in (0, 1).$$

Recall $\bar{x}_0 = \frac{1}{n} \mathbf{1}' \mathbf{x}_0$, $\mathbf{d} = \Pi \mathbf{x}_0$ so that $\mathbf{u}' \mathbf{x}_0 = \sqrt{n} \bar{x}_0$. Since

$$\tilde{A}(\mathbf{u}) = a_1 \mathbf{u}\mathbf{u}' + A_\perp,$$

we have

$$\Pi \tilde{A}(\mathbf{u})^t \mathbf{x}_0 = A_\perp^t \mathbf{d}, \quad \mathbf{x}_0' (\tilde{A}(\mathbf{u})')^t \mathbf{u} = a_1^t \sqrt{n} \bar{x}_0.$$

Hence

$$\Pi Z \mathbf{u} = \sqrt{n} \bar{x}_0 (I - \delta a_1 A_\perp)^{-1} \mathbf{d}.$$

Because A is doubly stochastic, $\|A\|_2 \leq \sqrt{\|A\|_1 \|A\|_\infty} = 1$. Hence $\rho(A_\perp) \leq \|A_\perp\|_2 \leq 1$. Since $\delta a_1 < 1$, the matrix $I - \delta a_1 A_\perp$ is invertible. If uniform exposure were optimal, Proposition 1 would imply that it satisfies (7). Projecting the right-hand side of (7) onto $\mathbf{u}^\perp$ gives

$$\Pi K^* \tilde{A}(\mathbf{u}) Z \mathbf{p} = \sqrt{n} \bar{x}_0 p_1 K_\perp A_\perp (I - \delta a_1 A_\perp)^{-1} \mathbf{d}.$$

Since $K_\perp$ and $I - \delta a_1 A_\perp$ are invertible on $\mathbf{u}^\perp$, this expression is zero if and only if $A_\perp \mathbf{d} = A \mathbf{d} = \mathbf{0}$. Thus when $A \mathbf{d} \neq \mathbf{0}$, its right-hand side has a nonzero disagreement component, whereas $\mathbf{b} = \mathbf{u}$ has none. Uniform exposure therefore cannot be optimal. $\square$

Proof of Proposition 3: By Lemma 2, an optimal exposure rule satisfies $\|\mathbf{b}\| = 1$. Fix such a rule $\mathbf{b}$. Since $A = \mathbf{1} \mathbf{a}'$, the Riccati equation implies that

$$K = I + \nu \mathbf{a} \mathbf{a}'$$

for some scalar $\nu > 0$. Hence $\mathbf{x}_0' K \mathbf{x}_0 = \|\mathbf{x}_0\|^2 + \nu (\mathbf{a}' \mathbf{x}_0)^2$. For every $\mathbf{x}_0$ with $\mathbf{a}' \mathbf{x}_0 \neq 0$, the optimal exposure rule therefore minimizes the same scalar ν and is independent of $\mathbf{x}_0$. Substituting $K = I + \nu \mathbf{a} \mathbf{a}'$ into the Riccati equation gives

$$\nu = \delta \left[n + \nu - \delta \frac{((\mathbf{1} + \nu \mathbf{a})' \mathbf{b})^2}{\mathbf{b}'((c + \delta)I + \delta \nu \mathbf{a} \mathbf{a}') \mathbf{b}} \right].$$

At an optimum, $\nu(\mathbf{b})$ is minimized over the unit sphere. Hence, for every tangent direction $\mathbf{h}$ satisfying $(\mathbf{b}^*)' \mathbf{h} = 0$, the directional derivative of $\nu(\mathbf{b})$ at $\mathbf{b}^*$ vanishes. Write the scalar Riccati equation as

$$F(\mathbf{b}, \nu) \equiv (1 - \delta)\nu - \delta n + \delta^2 R(\mathbf{b}, \nu) = 0, \quad \text{where } R(\mathbf{b}, \nu) = \frac{((\mathbf{1} + \nu \mathbf{a})' \mathbf{b})^2}{\mathbf{b}'((c + \delta)I + \delta \nu \mathbf{a} \mathbf{a}') \mathbf{b}}.$$

Differentiating $F(\mathbf{b}, \nu(\mathbf{b})) = 0$ in a tangent direction at $\mathbf{b}^*$ and using the vanishing derivative

of $\nu(\mathbf{b})$ gives

$$D_{\mathbf{b}}R(\mathbf{b}^*, \nu^*)[\mathbf{h}] = 0.$$

Thus, for its associated value ν^* , $\mathbf{b}^*$ is a constrained critical point of $R(\mathbf{b}, \nu^*)$. For any positive definite matrix M and nonzero vector $\mathbf{y}$, the constrained critical points of $\frac{(\mathbf{y}'\mathbf{b})^2}{\mathbf{b}'M\mathbf{b}}$ either satisfy $\mathbf{y}'\mathbf{b} = 0$ or are proportional to $M^{-1}\mathbf{y}$. The former are global minima and the latter are global maxima. In the present problem, $M = (c + \delta)I + \delta\nu\mathbf{a}\mathbf{a}'$ and $\mathbf{y} = \mathbf{1} + \nu\mathbf{a}$. The zero-numerator case implies $A'K\mathbf{b} = \mathbf{0}$, so the optimal intervention is identically zero. Since $\mathbf{a}'\mathbf{x}_0 \neq 0$, Lemma 2 rules out this case. Hence

$$\mathbf{b}^* \propto ((c + \delta)I + \delta\nu\mathbf{a}\mathbf{a}')^{-1}(\mathbf{1} + \nu\mathbf{a}).$$

By the Sherman–Morrison formula,

$$\mathbf{b}^* \propto \mathbf{1} + \frac{\nu c}{c + \delta + \delta\nu\|\mathbf{a}\|^2}\mathbf{a}.$$

The coefficient on $\mathbf{a}$ is strictly positive. Since $\mathbf{b}$ and $-\mathbf{b}$ are equivalent exposure rules, choose the normalization for which the coefficient on $\mathbf{1}$ is positive. Then

$$\mathbf{b}^* = \eta\mathbf{1} + \kappa\mathbf{a}, \quad \eta > 0, \quad \kappa > 0.$$

Consequently, $b_i^* - b_j^* = \kappa(a_i - a_j)$ and the conclusion follows. $\square$

Proof of Proposition 4: Work in the eigenbasis of A_0 and let $P = (\mathbf{p}_1, \dots, \mathbf{p}_n)$ where $\mathbf{p}_1 = \mathbf{u}$. Thus $\mathbf{e}_j$ denotes the j th coordinate vector in the transformed basis. Let $\tilde{A} = P'\tilde{A}P$ denote the induced opinion-transition matrix in the eigenbasis. Then $\tilde{A}_0 = \text{diag}(\tau, \lambda_2, \dots, \lambda_n)$. By scale invariance, any initial opinion near $\mathbf{x}_0 \propto \mathbf{1}$ can be normalized as

$$\tilde{\mathbf{x}}_0 = \mathbf{e}_1 + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \perp \mathbf{e}_1.$$

Parameterize unit exposure rules near $\mathbf{e}_1$ by

$$\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \sqrt{1 - \|\boldsymbol{\zeta}\|^2} \mathbf{e}_1 + \boldsymbol{\zeta}, \quad \boldsymbol{\zeta} \perp \mathbf{e}_1.$$

Hence

$$\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \mathbf{e}_1 + \boldsymbol{\zeta} + O(\|\boldsymbol{\zeta}\|^2).$$

At the benchmark $(\zeta, \xi, \varepsilon) = (\mathbf{0}, \mathbf{0}, 0)$, the Riccati matrix $\tilde{K}_0 = \text{diag}(\tilde{k}_1, \dots, \tilde{k}_n)$. The closed-loop matrix is $\tilde{A}_0 = \text{diag}(\tau, \lambda_2, \dots, \lambda_n)$. Moreover, $\tilde{\mathbf{p}}_0 = \ell_1 \mathbf{e}_1$ where

$$\ell_1 = \frac{\delta \tilde{k}_1}{c + \delta \tilde{k}_1}, \quad \tau = 1 - \ell_1 = \frac{c}{c + \delta \tilde{k}_1} \in (0, 1),$$

and, for $j > 1$, $\tilde{k}_j = \frac{1}{1 - \delta \lambda_j^2}$. Since (δ, c) are fixed, the average-coordinate equations determine τ , $\tilde{k}_1$, and ℓ_1 . Hence the coefficients below depend on j only through λ_j . Also,

$$\tilde{Z}_0 = z_1 \mathbf{e}_1 \mathbf{e}_1', \quad z_1 = \frac{1}{1 - \delta \tau^2}.$$

We first compute the tangent Hessian common to both perturbations. For each $j > 1$, consider the one-dimensional exposure rotation

$$\tilde{\mathbf{b}}_j(\gamma) = \sqrt{1 - \gamma^2} \mathbf{e}_1 + \gamma \mathbf{e}_j$$

and write $\tilde{K}_{1j}(\gamma) = q_j \gamma + o(\gamma)$ and $\tilde{K}_{11}(\gamma) = \tilde{k}_1 + h_j \gamma^2 + o(\gamma^2)$. Matching the coefficient of γ in the $(1, j)$ entry of the transformed Riccati equation gives q_j ; substituting this into the $(1, 1)$ entry and matching the coefficient of γ^2 gives h_j . Thus

$$q_j = -\frac{\delta \lambda_j \ell_1 \tilde{k}_j}{1 - \delta \tau \lambda_j} \quad \text{and} \quad h_j = \frac{\delta [\ell_1 \tau (\tilde{k}_1 - 2q_j) + \ell_1^2 \tilde{k}_j]}{1 - \delta \tau^2}.$$

Because $|\lambda_j| \leq 1$,

$$1 - \delta \tau \lambda_j > 0, \quad 1 + \delta \tau \lambda_j > 0,$$

and

$$\tau(\tilde{k}_1 - 2q_j) + \ell_1 \tilde{k}_j = \frac{\tau \tilde{k}_1 (1 - \delta \tau \lambda_j) + \ell_1 \tilde{k}_j (1 + \delta \tau \lambda_j)}{1 - \delta \tau \lambda_j} > 0.$$

Hence $h_j > 0$ for every $j > 1$.

At the benchmark, reversing the sign of any disagreement coordinate leaves D , $\mathbf{e}_1$, and $Q = I$ unchanged. Hence value is invariant under $\zeta_j \mapsto -\zeta_j$ for every $j > 1$. It is therefore even in each ζ_j , which rules out all quadratic cross terms $\zeta_j \zeta_l$, $j \neq l$. The constrained Hessian of value with respect to ζ is consequently $-2 \text{diag}(h_2, \dots, h_n)$. Because $h_j > 0$ for every $j > 1$, this Hessian is negative definite, so $\zeta = \mathbf{0}$ is a nondegenerate strict local

maximum at the benchmark. Let

$$\Psi(\zeta; \xi, \varepsilon) = D_\zeta \tilde{v}(\mathbf{e}_1 + \xi; \tilde{\mathbf{b}}(\zeta), D + \varepsilon \tilde{B})$$

denote the first-order condition in the local coordinates. This map is continuously differentiable, satisfies $\Psi(\mathbf{0}; \mathbf{0}, 0) = \mathbf{0}$, and has

$$D_\zeta \Psi(\mathbf{0}; \mathbf{0}, 0) = -2 \operatorname{diag}(h_2, \dots, h_n),$$

which is nonsingular. The implicit-function theorem therefore yields a unique continuously differentiable branch $\zeta^*(\xi, \varepsilon)$ near $(\mathbf{0}, 0)$, with $\zeta^*(\mathbf{0}, 0) = \mathbf{0}$. The next two calculations give its partial derivatives at the benchmark; the joint expansion follows from its first-order Taylor approximation.

We now differentiate the local optimum with respect to initial disagreement, holding $\varepsilon = 0$. The terms in the local loss that depend on ζ are

$$\sum_{j>1} (h_j \zeta_j^2 + 2q_j \zeta_j \xi_j) + o(\|\zeta\|^2 + \|\zeta\| \|\xi\|).$$

The first-order condition in direction j is therefore

$$h_j \zeta_j + q_j \xi_j + o(\|\zeta\| + \|\xi\|) = 0.$$

Since $h_j > 0$,

$$\frac{\partial \zeta_j^*}{\partial \xi_j}(\mathbf{0}, 0) = -\frac{q_j}{h_j} \equiv M(\lambda_j),$$

while

$$\frac{\partial \zeta_j^*}{\partial \xi_l}(\mathbf{0}, 0) = 0 \quad \text{for } l \neq j.$$

Let $\tilde{k}(\lambda) = 1/(1 - \delta\lambda^2)$. Substituting the expressions for q_j and h_j gives

$$M(\lambda) = \frac{\lambda \tilde{k}(\lambda)(1 - \delta\tau^2)}{\tau \tilde{k}_1(1 - \delta\tau\lambda) + \ell_1 \tilde{k}(\lambda)(1 + \delta\tau\lambda)}.$$

The average-coordinate Riccati equation gives

$$\tilde{k}_1 = \frac{1}{1 - \delta\tau}, \quad \ell_1 = 1 - \tau.$$

The denominator in the expression for $M(\lambda)$ is positive on $[-1, 1]$, so $M(\lambda)$ has the sign of λ . Multiplying numerator and denominator by $1 - \delta\lambda^2$, write

$$M(\lambda) = \frac{\lambda(1 - \delta\tau^2)}{N(\lambda)}, \quad N(\lambda) = \tau\tilde{k}_1(1 - \delta\tau\lambda)(1 - \delta\lambda^2) + (1 - \tau)(1 + \delta\tau\lambda).$$

Then

$$(1 - \delta\tau)(N(\lambda) - \lambda N'(\lambda)) = 1 - \delta\tau + \delta\tau^2 + \delta\tau\lambda^2 - 2\delta^2\tau^2\lambda^3 > 0.$$

For $\lambda \leq 0$ positivity is immediate. On $[0, 1]$, its derivative is $2\delta\tau\lambda(1 - 3\delta\tau\lambda)$, so any interior critical point is a maximum; the expression is strictly positive at both endpoints. Hence $M'(\lambda) > 0$ on $[-1, 1]$. Moreover, direct simplification gives, for $\lambda \in [0, 1]$,

$$\lambda - M(\lambda) = \frac{\delta\lambda\tau(1 - \lambda)[\lambda + 2\tau - \delta\tau(\lambda^2 + \lambda + \tau)]}{(1 - \delta\lambda^2)(1 - \delta\tau)\mathcal{D}(\lambda)},$$

where $\mathcal{D}(\lambda) > 0$ is the denominator in the expression for $M(\lambda)$. The term in brackets is concave in λ and strictly positive at both endpoints. Hence $M(0) = 0$, $M(1) = 1$ and $0 < M(\lambda) < \lambda$ for $\lambda \in (0, 1)$.

We next differentiate with respect to the network perturbation, holding $\xi = 0$. Let

$$\tilde{B} = P'BP, \quad \tilde{\mathbf{s}} = \tilde{B}'\mathbf{e}_1 = P'\mathbf{s}, \quad \mathbf{s} = B'\mathbf{u}.$$

Because $B\mathbf{1} = \mathbf{0}$, $\tilde{B}\mathbf{e}_1 = \mathbf{0}$ and $\mathbf{e}_1'\tilde{\mathbf{s}} = 0$. Holding $\tilde{\mathbf{b}} = \mathbf{e}_1$, define

$$\dot{\tilde{K}} = \left. \frac{\partial \tilde{K}^*(\mathbf{e}_1; D + \varepsilon\tilde{B})}{\partial \varepsilon} \right|_{\varepsilon=0}$$

and let

$$\tilde{\mathbf{k}} = (I - \mathbf{e}_1\mathbf{e}_1')\dot{\tilde{K}}\mathbf{e}_1.$$

Differentiating the transformed Riccati equation at the benchmark and projecting its first column onto $\mathbf{e}_1^\perp$ gives

$$\tilde{\mathbf{k}} = \delta\tau\tilde{k}_1\tilde{B}'\mathbf{e}_1 + \delta\tau D\tilde{\mathbf{k}},$$

which gives $\tilde{\mathbf{k}} = \delta\tau\tilde{k}_1(I - \delta\tau D)^{-1}\tilde{B}'\mathbf{e}_1$. Define

$$\mathbf{g} = (I - \delta\tau A'_0)^{-1}\mathbf{s} = \sum_{j>1} g_j \mathbf{p}_j.$$

Because $\mathbf{e}'_1\tilde{\mathbf{s}} = 0$ and D preserves $\mathbf{e}_1^\perp$, we also have $g_1 = 0$. Since

$$g_j = \frac{\tilde{s}_j}{1 - \delta\tau\lambda_j}, \quad \tilde{s}_j = \mathbf{e}'_j\tilde{B}'\mathbf{e}_1,$$

we obtain, for every $j > 1$,

$$\dot{K}_{j1} = \delta\tau\tilde{k}_1g_j.$$

We now differentiate the targeting condition. In the eigenbasis, the value gradient with respect to the exposure rule is

$$\nabla_{\tilde{\mathbf{b}}}\tilde{v} = 2\delta\tilde{K}\tilde{A}\tilde{Z}\tilde{\mathbf{p}}.$$

At the benchmark, this vector lies in $\text{span}\{\mathbf{e}_1\}$. Holding $\tilde{\mathbf{b}} = \mathbf{e}_1$, differentiation with respect to ε gives, for each $j > 1$,

$$\left. \frac{\partial}{\partial \varepsilon} \mathbf{e}'_j \nabla_{\tilde{\mathbf{b}}} \tilde{v} \right|_{(\boldsymbol{\xi}, \varepsilon) = (\mathbf{0}, 0)} = 2\delta\tau z_1 \ell_1 \dot{K}_{j1}.$$

Since $\tilde{B}\mathbf{e}_1 = \mathbf{0}$, the remaining benchmark factors map $\mathbf{e}_1$ into $\text{span}\{\mathbf{e}_1\}$. Hence the terms involving the derivatives of $\tilde{A}$, $\tilde{Z}$, and $\tilde{\mathbf{p}}$ remain in $\text{span}\{\mathbf{e}_1\}$ and vanish after projection onto $\mathbf{e}_j$. The change in the exposure rule contributes through the tangent Hessian. Differentiating the projected first-order condition along the local optimum therefore gives

$$-2h_j \frac{\partial \zeta_j^*}{\partial \varepsilon}(\mathbf{0}, 0) + 2\delta\tau z_1 \ell_1 \dot{K}_{j1} = 0.$$

Using $\dot{K}_{j1} = \delta\tau\tilde{k}_1g_j$ and $z_1 = 1/(1 - \delta\tau^2)$, we obtain

$$\frac{\partial \zeta_j^*}{\partial \varepsilon}(\mathbf{0}, 0) = \chi(\lambda_j)g_j, \quad \text{where} \quad \chi(\lambda_j) = \frac{\delta^2\tau^2\ell_1\tilde{k}_1}{(1 - \delta\tau^2)h_j} > 0.$$

Taking the total differential of the jointly differentiable branch gives, for each $j > 1$,

$$\zeta_j^*(\boldsymbol{\xi}, \varepsilon) = M(\lambda_j)\xi_j + \varepsilon\chi(\lambda_j)g_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Since $\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \mathbf{e}_1 + \boldsymbol{\zeta} + O(\|\boldsymbol{\zeta}\|^2)$, it follows that

$$\tilde{\mathbf{b}}^*(\boldsymbol{\xi}, \varepsilon) = \mathbf{e}_1 + \sum_{j>1} [M(\lambda_j)\xi_j + \varepsilon\chi(\lambda_j)g_j] \mathbf{e}_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Thus persistence and leverage add to first order; their interaction enters only at higher order.

Finally, we verify the clique claim following Proposition 4. Let $A_0 = J/n$ and consider the perturbation

$$B = \mathbf{1} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right)'.$$

Then

$$\mathbf{s} = B' \mathbf{u} = \sqrt{n} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right).$$

Because $\mathbf{s} \perp \mathbf{1}$ and $A_0' \mathbf{s} = \mathbf{0}$,

$$\mathbf{g} = (I - \delta\tau A_0')^{-1} \mathbf{s} = \mathbf{s}.$$

Moreover, every disagreement eigenvalue of A_0 is zero, so $\chi(\lambda_j) = \chi(0) > 0$ for every $j > 1$. Setting $\boldsymbol{\xi} = \mathbf{0}$ and returning to the original coordinates therefore gives

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon\chi\sqrt{n} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right) + o(|\varepsilon|).$$

Equivalently, for some $C > 0$,

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon C \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right) + o(|\varepsilon|).$$

Hence $b_i^*(\varepsilon) - b_j^*(\varepsilon) = \varepsilon C(a_i - a_j) + o(|\varepsilon|)$, which recovers locally the pairwise authority pattern in Proposition 3. $\square$

A.3 Proofs from Section 5

Lemma 4. *Let $\mathbf{b} \in S^{n-1}$ be any coarse exposure rule. The optimality gap from coarse targeting under $\mathbf{b}$ to full targeting can be expressed as*

$$G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}'^t \Phi(\mathbf{b}) \tilde{A}^t, \quad \Delta(\mathbf{b}) = \mathbf{x}_0' G(\mathbf{b}) \mathbf{x}_0, \quad (10)$$

where $\Phi(\mathbf{b}) \equiv (\mathbf{b}\mathbf{p}' - F^f)'(cI + \delta K^f)(\mathbf{b}\mathbf{p}' - F^f) \geq 0$ is the gap source matrix.

Proof of Proposition 5: By Lemma 4, $G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t$ and $\Delta(\mathbf{b}) = \mathbf{x}_0' G(\mathbf{b}) \mathbf{x}_0$. Therefore,

$$\Delta(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_0' \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t \mathbf{x}_0 = \sum_{t=0}^{\infty} \delta^t \text{tr} \left(\Phi(\mathbf{b}) \tilde{A}^t \mathbf{x}_0 \mathbf{x}_0' \tilde{A}^{t'} \right) = \text{tr} \left(\Phi(\mathbf{b}) \sum_{t=0}^{\infty} \delta^t \tilde{A}^t \mathbf{x}_0 \mathbf{x}_0' \tilde{A}^{t'} \right).$$

By definition, $Z(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}^t \mathbf{x}_0 \mathbf{x}_0' \tilde{A}^{t'}$. Thus $\Delta(\mathbf{b}) = \text{tr}(\Phi(\mathbf{b})Z(\mathbf{b}))$. $\square$

A.4 Proofs from Section 7

Proof of Proposition 6: We first compare myopic and full targeting. Matrix $\tilde{K}^f$ is the unique diagonal pd matrix solution to

$$\tilde{K}^f = I + \delta c D (\delta \tilde{K}^f + cI)^{-1} \tilde{K}^f D. \quad (11)$$

Matrix $\tilde{K}^f$ captures the decision-maker's future discounted loss, net of the cost of the optimal policy. Let $\tilde{K}_{jj}^f$ be the j th diagonal entry, which strictly increases in $|\lambda_j|$. Let $\Sigma = \sigma^2 I$. Since the eigenbasis transformation is orthogonal, the shock covariance remains Σ in the transformed coordinates. The decision-maker's value consists of two parts:

$$\tilde{v}^f(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^f \tilde{\mathbf{x}}_0 - \sum_{t=0}^{\infty} \delta^{t+1} \mathbb{E} \left(\tilde{\nu}_t' \tilde{K}^f \tilde{\nu}_t \right) = -\tilde{\mathbf{x}}_0' \tilde{K}^f \tilde{\mathbf{x}}_0 - \frac{\delta}{1-\delta} \text{tr}(\tilde{K}^f \Sigma). \quad (12)$$

For myopic targeting, without noise, the value is

$$-\tilde{\mathbf{x}}_0' \left(I - \frac{\delta c^2}{(\delta + c)^2} D^2 \right)^{-1} \left(I + \frac{c \delta^2}{(\delta + c)^2} D^2 \right) \tilde{\mathbf{x}}_0 = -\tilde{\mathbf{x}}_0' \tilde{K}^m \tilde{\mathbf{x}}_0, \quad (13)$$

where $\tilde{K}^m$ is the product of two matrices above, the first captures how each $\tilde{\mathbf{x}}_t$ depends on $\tilde{\mathbf{x}}_0$ given the optimal policy, and the second is how each stage payoff depends on $\tilde{\mathbf{x}}_t$. Clearly, $\tilde{K}^m$ is diagonal with entry $\tilde{K}_{jj}^m$, which strictly increases in $|\lambda_j|$. Each eigen-dimension is an independent scalar linear-quadratic problem. Full targeting uses the unique dynamically optimal scalar feedback, whereas the myopic feedback differs from it whenever $\lambda_j \neq 0$. Strict convexity of the scalar Bellman objective therefore implies $\tilde{k}_{jj}^f < \tilde{k}_{jj}^m$ whenever $\lambda_j \neq 0$, with equality when $\lambda_j = 0$. Moreover, the scalar Riccati equation implies that $\tilde{k}_{jj}^f$ is

increasing in λ_j^2 . Since $|\lambda_j| \leq 1 = \lambda_1$, we have $\tilde{k}_{jj}^f \leq \tilde{k}_{11}^f$. With noise, we have

$$\tilde{v}^m(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^m \tilde{\mathbf{x}}_0 - \frac{\delta}{1-\delta} \text{tr}(\tilde{K}^m \Sigma).$$

Let

$$\alpha = \frac{\delta \sigma^2}{1-\delta}, \quad \text{and} \quad \eta_j = \frac{\tilde{k}_{jj}^m - \tilde{k}_{jj}^f}{\tilde{k}_{jj}^f} \geq 0$$

for each j . The inequality is strict whenever $\lambda_j \neq 0$. Since

$$|\tilde{v}^f| = \sum_{j=1}^n ((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f \quad \text{and} \quad |\tilde{v}^m| = \sum_{j=1}^n ((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f (1 + \eta_j),$$

we can write

$$\frac{\tilde{v}^f}{\tilde{v}^m} = \frac{1}{1 + \sum_{j=1}^n \pi_j \eta_j},$$

where

$$\pi_j = \frac{((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f}{\sum_{l=1}^n ((\tilde{x}_0^l)^2 + \alpha) \tilde{k}_{ll}^f}, \quad \sum_{j=1}^n \pi_j = 1.$$

Let $\bar{\eta} = \max_j \eta_j$. Because A is row stochastic, $\lambda_1 = 1$, so $\eta_1 > 0$. Moreover, $\tilde{k}_{jj}^f \leq \tilde{k}_{11}^f$ for every j , and therefore

$$\pi_1 \geq \frac{\alpha}{\|\tilde{\mathbf{x}}_0\|^2 + n\alpha}.$$

Define $\underline{\eta} = \frac{\alpha}{\|\tilde{\mathbf{x}}_0\|^2 + n\alpha} \eta_1 > 0$. It follows that

$$\underline{\eta} \leq \sum_{j=1}^n \pi_j \eta_j \leq \bar{\eta}, \quad \text{or} \quad \frac{1}{1 + \bar{\eta}} \leq \left| \frac{\tilde{v}^f}{\tilde{v}^m} \right| \leq \frac{1}{1 + \underline{\eta}}.$$

In particular, full targeting strictly dominates myopic targeting.

Second, we compare full and coarse targeting. Suppose first that $\text{rank}(A) = 1$. Since A is symmetric and row stochastic, $A = J/n$. Hence, in the eigenbasis of A , $D = \text{diag}(1, 0, \dots, 0)$. Under full targeting, the optimal feedback matrix is diagonal and has zero entries in every disagreement dimension. Thus full targeting intervenes only in the average-opinion dimension. The uniform coarse exposure rule $\tilde{\mathbf{b}} = \mathbf{e}_1$ implements exactly the same intervention. In the first dimension, the coarse and full Riccati equations coincide, while for every $j > 1$, $\tilde{k}_{jj}^c = \tilde{k}_{jj}^f = 1$. Therefore $\tilde{K}^c = \tilde{K}^f$ and $\tilde{v}^c(\tilde{\mathbf{x}}_0) = \tilde{v}^f(\tilde{\mathbf{x}}_0)$. Now

suppose $\text{rank}(A) > 1$. Full targeting weakly dominates coarse targeting because every coarse intervention $\tilde{\mathbf{b}} r_t$ is feasible under full targeting at the same cost. Under full targeting,

$$\tilde{\mathbf{u}}_t^f = -\tilde{F}^f \tilde{\mathbf{x}}_t, \quad \tilde{F}^f = \delta(cI + \delta \tilde{K}^f)^{-1} \tilde{K}^f D.$$

Since $\tilde{K}^f$ is pd and diagonal, $\text{rank}(\tilde{F}^f) = \text{rank}(D) = \text{rank}(A) > 1$. Thus the full-targeting intervention does not always lie in any fixed one-dimensional subspace. By contrast, under any coarse exposure rule $\tilde{\mathbf{b}}$, every intervention lies in $\text{span}\{\tilde{\mathbf{b}}\}$. For a fixed $\tilde{\mathbf{b}}$, coarse targeting can therefore match the full-targeting intervention only at states satisfying $\tilde{F}^f \tilde{\mathbf{x}}_t \in \text{span}\{\tilde{\mathbf{b}}\}$. Because $\text{rank}(\tilde{F}^f) > 1$, this set is a proper subspace of $\mathbb{R}^n$. Since $\sigma > 0$, the state has full support, so it lies outside this subspace with positive probability. At those states, no scalar r_t can reproduce the full-targeting intervention. Since the Bellman objective is strictly concave in the intervention vector, coarse targeting then yields a strictly lower payoff. Thus full targeting strictly dominates coarse targeting whenever $\text{rank}(A) > 1$. $\square$

Proof of Proposition 7: We show each of the two claims in turn.

Step 1: An optimal coarse exposure rule can be chosen in $\text{span}\{\mathbf{u}, \mathbf{d}\}$. Choose the eigenbasis so that its second vector is aligned with $\mathbf{d}$. Let

$$d = \|\mathbf{d}\|, \quad P = \left(\mathbf{u}, \frac{\mathbf{d}}{d}, \mathbf{p}_3, \dots, \mathbf{p}_n \right).$$

Then $\tilde{\mathbf{x}}_0 = P' \mathbf{x}_0 = (\sqrt{n} \bar{x}_0, d, 0, \dots, 0)'$ and $D = \text{diag}(1, \rho, \dots, \rho)$. Take any optimal exposure rule and write $\tilde{\mathbf{b}} = a\mathbf{e}_1 + b\mathbf{g}$, $\mathbf{g} \perp \mathbf{e}_1$ and $\|\mathbf{g}\| = 1$, where $b \geq 0$. If $b = 0$, the claim is true. Fixing a, b , the cost matrix has the block form

$$\tilde{K}^c(\mathbf{g}) = \begin{pmatrix} \bar{K}^c(a, b) & 0 \\ 0 & \tilde{k}_\rho^\emptyset I_{n-2} \end{pmatrix}$$

relative to the decomposition into $\text{span}\{\mathbf{e}_1, \mathbf{g}\}$ and its orthogonal complement. Because all disagreement directions have the same eigenvalue and shocks are isotropic, $\bar{K}^c(a, b)$ and $\text{tr } \tilde{K}^c(\mathbf{g})$ do not depend on the particular choice of $\mathbf{g}$.

Let $s = d\mathbf{e}_2' \mathbf{g}$. The initial disagreement orthogonal to $\mathbf{g}$ has squared norm $d^2 - s^2$, so, up to

terms independent of $\mathbf{g}$,

$$\tilde{v}^c(s) = -2\bar{K}_{12}^c(a, b)\sqrt{n}\bar{x}_0s - \left(\bar{K}_{22}^c(a, b) - \tilde{k}_\rho^\emptyset\right)s^2.$$

Because zero intervention is feasible, $\bar{K}_{22}^c(a, b) \leq \tilde{k}_\rho^\emptyset$. Hence $\tilde{v}^c(s)$ is convex on $[-d, d]$ and is maximized at $s = d$ or $s = -d$. Thus an optimal disagreement direction can be chosen as $\mathbf{g} = \mathbf{e}_2$ or $\mathbf{g} = -\mathbf{e}_2$. Therefore an optimal exposure rule can be chosen in $\text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$, or equivalently in $\text{span}\{\mathbf{u}, \mathbf{d}\}$.

Step 2: Asymptotic ranking. Consider a sequence of economies as $n \rightarrow \infty$. In the basis constructed in Step 1, $\tilde{\mathbf{x}}_0^{(n)} = \left(\mathbf{u}'_n \mathbf{x}_0^{(n)}, \|\mathbf{d}_n\|, 0, \dots, 0\right)'$. By assumption, $\|\tilde{\mathbf{x}}_0^{(n)}\| = O(1)$. For every disagreement dimension $j > 1$, the scalar cost coefficients under full targeting, myopic targeting, and no intervention are respectively $\tilde{k}_\rho^f$, $\tilde{k}_\rho^m$, and $\tilde{k}_\rho^\emptyset$. Then

$$\tilde{k}_\rho^f = 1 + \delta c \rho^2 \frac{\tilde{k}_\rho^f}{c + \delta \tilde{k}_\rho^f}, \quad \tilde{k}_\rho^m = \frac{1 + \frac{c\delta^2 \rho^2}{(c+\delta)^2}}{1 - \frac{\delta c^2 \rho^2}{(c+\delta)^2}}, \quad \tilde{k}_\rho^\emptyset = \frac{1}{1 - \delta \rho^2}.$$

Since $\rho \neq 0$, $0 < \tilde{k}_\rho^f < \tilde{k}_\rho^m < \tilde{k}_\rho^\emptyset$, because full targeting strictly improves on myopic targeting, and myopic targeting strictly improves on no intervention. Under full targeting,

$$-\tilde{v}_n^f = \tilde{k}_1^f (\mathbf{u}'_n \mathbf{x}_0^{(n)})^2 + \tilde{k}_\rho^f \|\mathbf{d}_n\|^2 + \frac{\delta \sigma^2}{1 - \delta} \left(\tilde{k}_1^f + (n-1)\tilde{k}_\rho^f \right).$$

Similarly, under myopic targeting,

$$-\tilde{v}_n^m = \tilde{k}_1^m (\mathbf{u}'_n \mathbf{x}_0^{(n)})^2 + \tilde{k}_\rho^m \|\mathbf{d}_n\|^2 + \frac{\delta \sigma^2}{1 - \delta} \left(\tilde{k}_1^m + (n-1)\tilde{k}_\rho^m \right).$$

The initial-state terms remain bounded by assumption. For coarse targeting, since the optimal exposure rule has support only on the first two coordinates,

$$\tilde{K}_n^c = \begin{pmatrix} \bar{K}_n^c & 0 \\ 0 & \tilde{k}_\rho^\emptyset I_{n-2} \end{pmatrix},$$

where $\bar{K}_n^c$ is the leading 2×2 block. Then

$$-\tilde{v}_n^c = \begin{pmatrix} \mathbf{u}'_n \mathbf{x}_0^{(n)} \\ \|\mathbf{d}_n\| \end{pmatrix}' \bar{K}_n^c \begin{pmatrix} \mathbf{u}'_n \mathbf{x}_0^{(n)} \\ \|\mathbf{d}_n\| \end{pmatrix} + \frac{\delta \sigma^2}{1 - \delta} \left(\text{tr } \bar{K}_n^c + (n-2)\tilde{k}_\rho^\emptyset \right).$$

Because zero intervention is feasible,

$$0 \prec \bar{K}_n^c \preceq \begin{pmatrix} \tilde{k}_1^\emptyset & 0 \\ 0 & \tilde{k}_\rho^\emptyset \end{pmatrix}.$$

Hence $\bar{K}_n^c$ is uniformly bounded in n . Since $\mathbf{u}'_n \mathbf{x}_0^{(n)} = O(1)$ and $\|\mathbf{d}_n\| = O(1)$,

$$-\tilde{v}_n^c = O(1) + \frac{\delta\sigma^2}{1-\delta} \left(\text{tr } \bar{K}_n^c + (n-2)\tilde{k}_\rho^\emptyset \right).$$

Therefore, dividing each value by n ,

$$\lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^m} = \frac{\tilde{k}_\rho^f}{\tilde{k}_\rho^m}, \quad \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^c} = \frac{\tilde{k}_\rho^f}{\tilde{k}_\rho^\emptyset}.$$

Since $0 < \tilde{k}_\rho^f < \tilde{k}_\rho^m < \tilde{k}_\rho^\emptyset$, the conclusion follows. □

Supplementary Appendix

S1 Additional proofs

Proof of Lemma 1: Fix $\mathbf{b} \neq \mathbf{0}$, $c > 0$, and $\delta \in (0, 1)$. We verify that the discounted linear-quadratic problem has a unique stabilizing quadratic value function. First, the no-intervention policy is feasible and yields a finite value. Under $r_t \equiv 0$,

$$\sum_{t=0}^{\infty} \delta^t \mathbf{x}'_t Q \mathbf{x}_t = \mathbf{x}' \left(\sum_{t=0}^{\infty} (\sqrt{\delta} A')^t Q (\sqrt{\delta} A)^t \right) \mathbf{x}.$$

Since A is row-stochastic, $\rho(A) \leq 1$, so $\rho(\sqrt{\delta} A) < 1$. The series therefore converges, and the value is finite for all initial opinions. To apply the standard discrete-time LQ theorem, write the problem in scaled variables

$$\mathbf{y}_t = (\sqrt{\delta})^t \mathbf{x}_t, \quad u_t = (\sqrt{\delta})^t r_t, \quad \text{then}$$

$$\mathbf{y}_{t+1} = \sqrt{\delta} A \mathbf{y}_t + \sqrt{\delta} \mathbf{b} u_t, \text{ and } \sum_{t=0}^{\infty} \delta^t (\mathbf{x}'_t Q \mathbf{x}_t + c r_t^2) = \sum_{t=0}^{\infty} (\mathbf{y}'_t Q \mathbf{y}_t + c u_t^2).$$

Because $\sqrt{\delta} A$ is Schur stable, the pair $(\sqrt{\delta} A, \sqrt{\delta} \mathbf{b})$ is stabilizable. Since $Q \succ 0$, the pair $(Q^{1/2}, \sqrt{\delta} A)$ is detectable. With $c > 0$, the standard discrete-time LQ theorem gives a unique stabilizing quadratic value function $v(\mathbf{x}) = -\mathbf{x}' K^* \mathbf{x}$, where $K^* \succ 0$, and a unique linear optimal policy. It remains only to record the associated equations. The Bellman equation in the original discounted variables is

$$v(\mathbf{x}) = \max_r \left\{ -\mathbf{x}' Q \mathbf{x} - c r^2 - \delta (A \mathbf{x} + \mathbf{b} r)' K^* (A \mathbf{x} + \mathbf{b} r) \right\}.$$

The first-order condition gives

$$r^*(\mathbf{x}) = -\delta (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \mathbf{b}' K^* A \mathbf{x} \equiv -\mathbf{p}' \mathbf{x}.$$

Substituting this rule into the Bellman equation and matching quadratic terms gives

$$K^* = Q + \delta A' \left(K^* - \delta K^* \mathbf{b} (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \mathbf{b}' K^* \right) A.$$

The stabilizing solution is unique and satisfies $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$, so the Riccati equation uniquely determines the continuation value and the optimal intervention rule. $\square$

Proof of Lemma 2: Fix $\mathbf{b} \neq \mathbf{0}$ and let $\gamma > 1$. The dynamics under exposure rule $\gamma\mathbf{b}$ and intervention level $\tilde{r}_t$ are identical to those under exposure rule $\mathbf{b}$ and intervention level $r_t = \gamma\tilde{r}_t$. Hence replacing $\mathbf{b}$ by $\gamma\mathbf{b}$ is equivalent to lowering the intervention cost from c to c/γ^2 . The value weakly increases, and strictly increases whenever the optimal intervention is nontrivial. It remains to show that intervention is strictly valuable. Let K^0 denote the no-intervention loss matrix,

$$K^0 = \sum_{t=0}^{\infty} \delta^t (A')^t Q A^t.$$

Since $Q A \mathbf{x}_0 \neq \mathbf{0}$, we have $A \mathbf{x}_0 \neq \mathbf{0}$, and therefore $K^0 A \mathbf{x}_0 \neq \mathbf{0}$. Choosing an exposure direction proportional to $-K^0 A \mathbf{x}_0$ and a sufficiently small positive intervention at date zero, followed by zero intervention thereafter, strictly improves on no intervention. Hence every optimal exposure rule induces a nontrivial intervention. Therefore, if $\|\mathbf{b}\| < 1$, scaling $\mathbf{b}$ up slightly while remaining feasible strictly improves the value. Every optimal exposure rule must satisfy $\|\mathbf{b}^*\| = 1$. $\square$

To conduct local comparative statics, we call a stationary point *nondegenerate* if the Hessian of the Lagrangian restricted to the tangent space of the unit sphere is nonsingular.

Corollary S1. *Suppose every stationary point of the outer problem (5) is nondegenerate. Then stationary points are isolated and vary smoothly with the primitives; there are finitely many of them, and every local maximum is strict.*

Proof of Corollary S1: Fix a value of the scalar primitive θ and a stationary pair $(\mathbf{b}^*, \mu^*)$. Let $T_{\mathbf{b}^*} = \{\mathbf{h} \in \mathbb{R}^n : \mathbf{b}^{*'} \mathbf{h} = 0\}$ be the tangent space of the unit sphere at $\mathbf{b}^*$. Let $P^* \equiv I - \mathbf{b}^* \mathbf{b}^{*'}$ denote the orthogonal projection onto $T_{\mathbf{b}^*}$. Define the constrained Hessian $H(\mathbf{b}^*, \theta) : T_{\mathbf{b}^*} \rightarrow T_{\mathbf{b}^*}$ by

$$H(\mathbf{b}^*, \theta) \mathbf{h} = P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h}, \quad \mathbf{h} \in T_{\mathbf{b}^*}.$$

Equivalently,

$$H(\mathbf{b}^*, \theta) = P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I)|_{T_{\mathbf{b}^*}}.$$

By nondegeneracy, this operator is nonsingular on $T_{\mathbf{b}^*}$. The bordered derivative of the first-order condition and the constraint is

$$\begin{pmatrix} v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I & -\mathbf{b}^* \\ \mathbf{b}^{*'} & 0 \end{pmatrix}.$$

To see that this matrix is nonsingular, suppose $(\mathbf{h}, a)$ lies in its kernel. Then

$$\begin{aligned} (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h} - a \mathbf{b}^* &= \mathbf{0}, \\ \mathbf{b}^{*'} \mathbf{h} &= 0. \end{aligned}$$

The second equation implies $\mathbf{h} \in T_{\mathbf{b}^*}$. Premultiplying the first equation by P^* gives $P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h} = \mathbf{0}$, because $P^* \mathbf{b}^* = \mathbf{0}$. By the definition of H , this is $H(\mathbf{b}^*, \theta) \mathbf{h} = \mathbf{0}$. Since $H(\mathbf{b}^*, \theta)$ is nonsingular on $T_{\mathbf{b}^*}$, $\mathbf{h} = \mathbf{0}$. The first equation then gives $a \mathbf{b}^* = \mathbf{0}$, and hence $a = 0$. The bordered derivative is therefore nonsingular. The implicit function theorem implies that each stationary exposure rule is locally unique and varies smoothly with the primitives. Nondegeneracy also implies that every stationary exposure rule is isolated. Since the stationary set is closed and contained in the compact unit sphere, it is finite. Finally, at a local maximum the constrained Hessian is negative semidefinite. Since it is nonsingular, it is negative definite, so every local maximum is strict. $\square$

Proof of Lemma 3: Fix the orthonormal eigenbasis used in the statement. Suppose by contradiction that a strict local optimum $\tilde{\mathbf{b}}^*$ has $\tilde{b}_j^* \neq 0$. For $\varepsilon \in (0, 1)$, define a nearby exposure rule $\hat{\mathbf{b}}$ by

$$\hat{b}_k = \tilde{b}_k^* \quad \text{for } k \neq j, \quad \hat{b}_j = (1 - \varepsilon) \tilde{b}_j^*.$$

Then $\|\hat{\mathbf{b}}\| < \|\tilde{\mathbf{b}}^*\| \leq 1$, so $\hat{\mathbf{b}}$ is feasible, distinct from $\tilde{\mathbf{b}}^*$, and arbitrarily close to it.

Fix any intervention sequence $\{r_t\}$. All coordinates $k \neq j$ follow the same path under $\tilde{\mathbf{b}}^*$ and $\hat{\mathbf{b}}$. In coordinate j , the two dynamics are

$$\tilde{x}_{t+1}^j = \lambda_j \tilde{x}_t^j + \tilde{b}_j^* r_t, \quad \hat{x}_{t+1}^j = \lambda_j \hat{x}_t^j + (1 - \varepsilon) \tilde{b}_j^* r_t.$$

Since $\lambda_j \tilde{x}_0^j = 0$, we have $\hat{x}_1^j = (1 - \varepsilon) \tilde{x}_1^j$. Induction then gives

$$\hat{x}_t^j = (1 - \varepsilon) \tilde{x}_t^j \quad \text{for every } t \geq 1.$$

Thus loss in coordinate j is weakly lower at every date, and strictly lower whenever $\tilde{x}_t^j \neq 0$ for some $t \geq 1$. Losses in all other coordinates and intervention costs are unchanged.

Therefore every intervention sequence available under $\tilde{\mathbf{b}}^*$ yields a weakly higher payoff under $\hat{\tilde{\mathbf{b}}}$. Maximizing over intervention sequences gives $\tilde{v}(\tilde{\mathbf{x}}_0; \hat{\tilde{\mathbf{b}}}) \geq \tilde{v}(\tilde{\mathbf{x}}_0; \tilde{\mathbf{b}}^*)$. Because $\hat{\tilde{\mathbf{b}}}$ is feasible, distinct from $\tilde{\mathbf{b}}^*$, and arbitrarily close to it, $\tilde{\mathbf{b}}^*$ cannot be a strict local maximum. Hence any strict local maximum must satisfy $\tilde{b}_j^* = 0$. $\square$

Proof of Corollary 1: If $\rho = 0$ and $\bar{x}_0 = 0$, then $A(\rho)\mathbf{x}_0 = \mathbf{0}$. Zero intervention is optimal for every exposure rule, so the value is constant in $\mathbf{b}$ and there is no strict local maximum. Otherwise, $A(\rho)\mathbf{x}_0 \neq \mathbf{0}$. Any strict local maximum must have unit norm. Indeed, if its optimal intervention is nontrivial, scaling the exposure rule outward slightly implements the same opinion path at strictly lower intervention cost. If its optimal intervention is identically zero, then zero intervention remains feasible under every nearby exposure rule, so it cannot be a strict local maximum.

Fix ρ . As in the text, write the initial opinion vector as $\mathbf{x}_0 = \sqrt{n}\bar{x}_0\mathbf{u} + \mathbf{d}$, where $\mathbf{u} = \mathbf{1}/\sqrt{n}$ and $\mathbf{d} \perp \mathbf{u}$. Consider any exposure rule with $\|\mathbf{b}\| = 1$. Decompose it into its average-opinion component and its disagreement component in a similar way. Let $\mathbf{b} = a\mathbf{u} + s\mathbf{h}$, where $\mathbf{h} \perp \mathbf{u}$, $\|\mathbf{h}\| = 1$, and $a^2 + s^2 = 1$. If $s = 0$, then $\mathbf{b} = a\mathbf{u} \in \text{span}\{\mathbf{u}, \mathbf{d}\}$, so there is nothing to prove. Suppose $s > 0$. Given the disagreement direction $\mathbf{h}$ chosen by the exposure rule, decompose initial disagreement as

$$\mathbf{d} = z\mathbf{h} + \mathbf{d}^\perp, \quad z = \mathbf{h}'\mathbf{d}, \quad \mathbf{d}^\perp \perp \mathbf{h}.$$

Then $\|\mathbf{d}^\perp\|^2 = \|\mathbf{d}\|^2 - z^2$. Since $A(\rho)\mathbf{u} = \mathbf{u}$ and $A(\rho)\mathbf{y} = \rho\mathbf{y}$ for every $\mathbf{y} \perp \mathbf{u}$, the subspace spanned by $\mathbf{u}$ and $\mathbf{h}$ is invariant. On this subspace, the controlled coordinates are $m_t = \mathbf{u}'\mathbf{x}_t$ and $y_t = \mathbf{h}'\mathbf{x}_t$, with dynamics $m_{t+1} = m_t + ar_t$ and $y_{t+1} = \rho y_t + sr_t$. All residual disagreement, orthogonal to both $\mathbf{u}$ and $\mathbf{h}$, receives no intervention and evolves autonomously with eigenvalue ρ . Thus $\mathbf{d}_{t+1}^\perp = \rho\mathbf{d}_t^\perp$.

Let $K_2(a, s)$ be the Riccati matrix for the two-dimensional controlled system, and let $k_{ij}(a, s)$ denote its ij -th entry. The residual disagreement has discounted loss coefficient

$$k_\rho = \sum_{t=0}^{\infty} \delta^t \rho^{2t} = \frac{1}{1 - \delta\rho^2}.$$

Hence the value generated by the exposure rule $\mathbf{b} = a\mathbf{u} + s\mathbf{h}$ is

$$v(\mathbf{x}_0; \mathbf{b}) = -k_{11}(a, s)n\bar{x}_0^2 - 2k_{12}(a, s)\sqrt{n}\bar{x}_0z - (k_{22}(a, s) - k_\rho)z^2 - k_\rho\|\mathbf{d}\|^2.$$

For fixed (a, s) , the only remaining choice is the disagreement direction $\mathbf{h}$, and this choice affects value only through $z = \mathbf{h}'\mathbf{d}$. Because the decision-maker can always choose zero intervention, the optimized loss from the controlled disagreement coordinate cannot exceed its uncontrolled loss. Hence $k_{22}(a, s) \leq k_\rho$, so $v(\mathbf{x}_0; \mathbf{b})$ is convex in z . If $|z| < \|\mathbf{d}\|$, then $\mathbf{h}$ can be rotated locally within the disagreement space so as to move z in either direction. Since value is convex in z , it cannot have a strict local maximum at such an interior value. Any strict local maximum must therefore satisfy $|z| = \|\mathbf{d}\|$. By equality in the Cauchy-Schwarz inequality, this holds if and only if $\mathbf{h} = \pm \frac{\mathbf{d}}{\|\mathbf{d}\|}$. Hence the disagreement component of $\mathbf{b}$ is proportional to $\mathbf{d}$, and therefore $\mathbf{b} \in \text{span}\{\mathbf{u}, \mathbf{d}\}$. $\square$

Proof of Lemma 4: Define the gap matrix $G(\mathbf{b}) \equiv K^*(\mathbf{b}) - K^f$. We first show that $G(\mathbf{b})$ satisfies the Lyapunov equation

$$G(\mathbf{b}) = \Phi(\mathbf{b}) + \delta\tilde{A}'G(\mathbf{b})\tilde{A},$$

and then establish a series representation.

Step 1: Riccati matrix identities. Recall that for any coarse exposure rule $\mathbf{b}$, we have $\mathbf{p} = \mathbf{p}(\mathbf{b})$ and $\tilde{A}(\mathbf{b}) = A - \mathbf{b}\mathbf{p}'$. Under the optimal scalar intervention level $r_t = -\mathbf{p}'\mathbf{x}_t$, the induced vector intervention is $\mathbf{u}_t = -\mathbf{b}\mathbf{p}'\mathbf{x}_t$, and the coarse transition matrix is $\tilde{A} = A - \mathbf{b}\mathbf{p}'$. The value iteration for the coarse problem can be rewritten as

$$K^*(\mathbf{b}) = Q + c\mathbf{p}\mathbf{p}' + \delta\tilde{A}'K^*(\mathbf{b})\tilde{A}. \quad (\text{S1})$$

The term $c\mathbf{p}\mathbf{p}'$ comes from the intervention cost

$$cr_t^2 = c(\mathbf{p}'\mathbf{x}_t)^2 = \mathbf{x}_t'c\mathbf{p}\mathbf{p}'\mathbf{x}_t.$$

Under full targeting, the optimal vector intervention is $\mathbf{u}_t = -F^f\mathbf{x}_t$, where

$$F^f = \delta(cI + \delta K^f)^{-1}K^fA.$$

The full-targeting value iteration satisfies

$$K^f = Q + c(F^f)'F^f + \delta(A - F^f)'K^f(A - F^f).$$

We use the following completion-of-squares identity. For any feedback matrix $F \in \mathbb{R}^{n \times n}$,

$$cF'F + \delta(A - F)'K^f(A - F) = K^f - Q + (F - F^f)'(cI + \delta K^f)(F - F^f). \quad (\text{S2})$$

To verify this identity, expand the right-hand side and use

$$(cI + \delta K^f)F^f = \delta K^f A, \quad (F^f)'(cI + \delta K^f) = \delta A'K^f,$$

together with the Riccati equation for K^f . Apply (S2) to the coarse feedback matrix $F^c(\mathbf{b}) = \mathbf{b}\mathbf{p}'$. Since $\|\mathbf{b}\| = 1$, we have $cF'F = c(\mathbf{b}\mathbf{p}')'(\mathbf{b}\mathbf{p}') = c\mathbf{p}\mathbf{b}'\mathbf{b}\mathbf{p}' = c\mathbf{p}\mathbf{p}'$. Hence

$$c\mathbf{p}\mathbf{p}' + \delta\tilde{A}'K^f\tilde{A} = K^f - Q + \Phi(\mathbf{b}), \quad (\text{S3})$$

where $\Phi(\mathbf{b}) \equiv (F^c(\mathbf{b}) - F^f)'(cI + \delta K^f)(F^c(\mathbf{b}) - F^f)$. Since $cI + \delta K^f > 0$, we have $\Phi(\mathbf{b}) \geq 0$.

Step 2: gap Lyapunov representation. Now return to the coarse identity (S1). Add and subtract $\delta\tilde{A}'K^f\tilde{A}$:

$$K^*(\mathbf{b}) = Q + \left[c\mathbf{p}\mathbf{p}' + \delta\tilde{A}'K^f\tilde{A} \right] + \delta\tilde{A}'(K^*(\mathbf{b}) - K^f)\tilde{A}.$$

Using (S3) and canceling Q gives

$$K^*(\mathbf{b}) = K^f + \Phi(\mathbf{b}) + \delta\tilde{A}'(K^*(\mathbf{b}) - K^f)\tilde{A}. \quad (\text{S4})$$

Therefore, with $G(\mathbf{b}) = K^*(\mathbf{b}) - K^f$,

$$G(\mathbf{b}) = \Phi(\mathbf{b}) + \delta\tilde{A}'G(\mathbf{b})\tilde{A}. \quad (\text{S5})$$

Step 3: Series representation. It remains to iterate (S5). Repeated substitution gives, for every $T \geq 0$,

$$G(\mathbf{b}) = \sum_{t=0}^T \delta^t \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t + \delta^{T+1} \tilde{A}^{(T+1)'} G(\mathbf{b}) \tilde{A}^{T+1}.$$

Because $\rho(\sqrt{\delta} \tilde{A}) < 1$, the final term converges to zero as $T \rightarrow \infty$. Hence

$$G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}' \Phi(\mathbf{b}) \tilde{A}^t.$$

Finally, $\Delta(\mathbf{b}) = v^f(\mathbf{x}_0) - v(\mathbf{x}_0; \mathbf{b}) = \mathbf{x}_0'(K^*(\mathbf{b}) - K^f)\mathbf{x}_0 = \mathbf{x}_0'G(\mathbf{b})\mathbf{x}_0$ as claimed. $\square$

S2 Persistent bias and coarse targeting

Suppose opinions evolve according to $\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t + \mathbf{w}$, where $\mathbf{w}$ is time invariant. Define the augmented state

$$\boldsymbol{\chi}_t = \begin{bmatrix} \mathbf{x}_t \\ \mathbf{w} \end{bmatrix}, \quad \bar{A} = \begin{bmatrix} A & I \\ \mathbf{0} & I \end{bmatrix}, \quad \bar{\mathbf{b}} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}, \quad \bar{Q} = \begin{bmatrix} Q & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}.$$

Then $\boldsymbol{\chi}_{t+1} = \bar{A}\boldsymbol{\chi}_t + \bar{\mathbf{b}}r_t$, and the decision-maker's problem becomes

$$\max_{\{r_t\}_{t \geq 0}} - \sum_{t=0}^{\infty} \delta^t (\boldsymbol{\chi}_t' \bar{Q} \boldsymbol{\chi}_t + cr_t^2). \quad (\text{S6})$$

For any fixed $\mathbf{b} \neq \mathbf{0}$, this is a discounted linear-quadratic problem in the augmented state. Its value has the form $v(\boldsymbol{\chi}; \mathbf{b}) = -\boldsymbol{\chi}' \bar{K}^*(\mathbf{b}) \boldsymbol{\chi}$, where $\bar{K}^*(\mathbf{b})$ is symmetric and positive semidefinite and solves the associated Riccati equation. The optimal intervention is

$$r_t^* = -\bar{\mathbf{p}}(\mathbf{b})' \boldsymbol{\chi}_t, \quad \text{or} \quad r_t^* = -\mathbf{p}_x(\mathbf{b})' \mathbf{x}_t - \mathbf{p}_w(\mathbf{b})' \mathbf{w}.$$

Let $v(\mathbf{b}; \mathbf{x}_0, \mathbf{w})$ denote the maximal value in (S6). Since $\boldsymbol{\chi}_0 = [\mathbf{x}_0' \ \mathbf{w}']'$, we can write

$$v(\mathbf{b}; \mathbf{x}_0, \mathbf{w}) = -\boldsymbol{\chi}_0' \bar{K}^*(\mathbf{b}) \boldsymbol{\chi}_0 = - \begin{bmatrix} \mathbf{x}_0 \\ \mathbf{w} \end{bmatrix}' \bar{K}^*(\mathbf{b}) \begin{bmatrix} \mathbf{x}_0 \\ \mathbf{w} \end{bmatrix}. \quad (\text{S7})$$

Corollary S2. *The augmented problem is well behaved: an optimal exposure rule exists, and the value depends locally smoothly on $\mathbf{b}$ and the primitives. For generic primitives, every optimum has unit norm and satisfies*

$$\mu \mathbf{b} = \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \bar{K}^* \tilde{A} Z_{\chi} \bar{\mathbf{p}}, \quad (\text{S8})$$

where $\tilde{A} = \bar{A} - \bar{\mathbf{b}}\bar{\mathbf{p}}'$ and $Z_\chi = \sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A})^t \chi_0 \chi_0' (\sqrt{\delta} \tilde{A}')^t$.

Although $\bar{Q}$ is only psd, the same argument applies because $\rho(\sqrt{\delta} \bar{A}) < 1$. Projecting the augmented exposure gradient onto its first n coordinates gives (S8), while the usual scaling argument gives unit norm whenever intervention is strictly valuable, which holds generically. The decision-maker's tradeoff now depends not only on the endogenous opinion path, but also on the interaction between opinions and the bias as summarized by Z_χ . Let

$$Z_\chi(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \chi_t \chi_t' = \begin{bmatrix} \sum_{t \geq 0} \delta^t \mathbf{x}_t \mathbf{x}_t' & \sum_{t \geq 0} \delta^t \mathbf{x}_t \mathbf{w}' \\ \sum_{t \geq 0} \delta^t \mathbf{w} \mathbf{x}_t' & \frac{1}{1 - \delta} \mathbf{w} \mathbf{w}' \end{bmatrix}.$$

The upper-left block records the endogenous opinion path, the lower-right block records the repeated presence of the bias, and the off-diagonal blocks record their interaction. Persistent bias therefore supplements rather than replaces the role of initial opinions.

Proposition S1. *Suppose A is symmetric and $Q = I$. Let $\Pi = I - \mathbf{u}\mathbf{u}'$, $\mathbf{d} = \Pi \mathbf{x}_0$, $\mathbf{w}_\perp = \Pi \mathbf{w}$, and define*

$$\mathcal{H}(\mathbf{x}_0, \mathbf{w}) = \text{span} \{ \mathbf{u}, A^{t+1} \mathbf{d}, A^t \mathbf{w}_\perp : t \geq 0 \}.$$

Then every strict local optimum satisfies $\mathbf{b}^ \in \mathcal{H}(\mathbf{x}_0, \mathbf{w})$.*

The subspace $\mathcal{H}(\mathbf{x}_0, \mathbf{w})$ contains three sources of payoff-relevant opinion movement: average opinion, disagreement inherited from $\mathbf{x}_0$ that survives at least one social update, and the network orbit of the disagreement repeatedly injected by $\mathbf{w}$. Exposure outside this subspace can only create additional opinion variation in directions containing neither inherited nor injected disagreement.

Proof of Proposition S1: Write $\mathcal{H} = \mathcal{H}(\mathbf{x}_0, \mathbf{w})$. Because A is symmetric, both $\mathcal{H}$ and $\mathcal{H}^\perp$ are invariant under A . Moreover, $A\mathbf{x}_0 + \mathbf{w} \in \mathcal{H}$. Thus, from date one onward, every component of opinions in $\mathcal{H}^\perp$ is generated only by intervention. Also, any strict local optimum must have unit norm. If its optimal intervention is nontrivial, scaling the exposure rule outward slightly implements the same opinion path at strictly lower intervention cost. If its optimal intervention is identically zero, every nearby exposure rule can reproduce the same payoff by also choosing zero intervention, so the optimum cannot be strict. Suppose

that $\mathbf{b}$ has a nonzero component in $\mathcal{H}^\perp$. Write

$$\mathbf{b} = q\mathbf{h} + s\mathbf{k}, \quad q \geq 0, \quad s > 0, \quad q^2 + s^2 = 1,$$

where $\mathbf{h} \in \mathcal{H}$ and $\mathbf{k} \in \mathcal{H}^\perp$ are unit vectors. If $q = 0$, choose any unit vector $\mathbf{h} \in \mathcal{H}$. Choose $q' > q$ sufficiently close to q , let

$$s' = \sqrt{1 - q'^2}, \quad \mathbf{b}' = q'\mathbf{h} + s'\mathbf{k},$$

and, for any intervention sequence $\{r_t\}$ under $\mathbf{b}$, define $r'_t = \frac{q}{q'}r_t$. The intervention generated inside $\mathcal{H}$ is unchanged: $q'\mathbf{h}r'_t = q\mathbf{h}r_t$. The intervention generated in $\mathcal{H}^\perp$ is reduced because

$$s'\mathbf{k}r'_t = \frac{s'q}{sq'}s\mathbf{k}r_t, \quad 0 \leq \frac{s'q}{sq'} < 1.$$

Since both subspaces are invariant under A , the opinion path inside $\mathcal{H}$ is unchanged, while the component in $\mathcal{H}^\perp$ and its associated loss are weakly reduced from date one onward. Intervention costs are also weakly lower: $(r'_t)^2 = \left(\frac{q}{q'}\right)^2 r_t^2 \leq r_t^2$, with strict inequality whenever $r_t \neq 0$. Hence, if intervention is nonzero at some date, $\mathbf{b}'$ yields a strictly higher payoff. If the optimal intervention under $\mathbf{b}$ is identically zero, choosing zero intervention under the nearby rule $\mathbf{b}'$ reproduces exactly the same opinion path and payoff. Hence $v(\mathbf{b}') \geq v(\mathbf{b})$, which already contradicts strict local optimality. Thus a strict local optimum cannot have a nonzero component in $\mathcal{H}^\perp$, and hence $\mathbf{b}^* \in \mathcal{H}(\mathbf{x}_0, \mathbf{w})$. $\square$

S3 Nonlinear updating and local disagreement dynamics

The effectiveness of coarse targeting depends on how disagreement propagates, rather than on the specific micro-foundation for opinion updating. For a smooth updating rule $\mathbf{x}_{t+1} = F(\mathbf{x}_t) + \mathbf{b}r_t$, consider local deviations around a no-intervention fixed point $\bar{\mathbf{x}}$, satisfying $F(\bar{\mathbf{x}}) = \bar{\mathbf{x}}$. Then

$$\Delta \mathbf{x}_{t+1} \approx J(\bar{\mathbf{x}})\Delta \mathbf{x}_t + \mathbf{b}r_t,$$

where $J(\bar{\mathbf{x}})$ is the Jacobian of F evaluated at $\bar{\mathbf{x}}$. Thus the relevant local object is the propagation matrix $J(\bar{\mathbf{x}})$. The baseline model is the special case in which this matrix is constant and equal to A . As an illustration, consider the Friedkin–Johnsen model (Friedkin

and Johnsen, 1990),

$$\mathbf{x}_{t+1} = (I - \Lambda)W\mathbf{x}_t + \Lambda\mathbf{x}_0 + \mathbf{b}r_t,$$

where W is row-stochastic and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_i \in [0, 1]$. Let $\bar{\mathbf{x}}$ be the steady state absent intervention:

$$\bar{\mathbf{x}} = (I - \Lambda)W\bar{\mathbf{x}} + \Lambda\mathbf{x}_0.$$

Defining deviations from this steady state by $\Delta\mathbf{x}_t = \mathbf{x}_t - \bar{\mathbf{x}}$, we obtain

$$\Delta\mathbf{x}_{t+1} = (I - \Lambda)W\Delta\mathbf{x}_t + \mathbf{b}r_t.$$

Hence the model takes the same local form as the baseline model, with $A = (I - \Lambda)W$. Here A is substochastic when some agents are partially stubborn. If stubbornness anchors every recurrent component of the network, then $\rho(A) < 1$, so deviations decay absent intervention. Components associated with eigenvalues close to one, however, decay slowly and therefore behave like persistent disagreement for a patient decision-maker. Since Friedkin–Johnsen updating is affine, this representation is exact rather than merely local.

The same local logic applies to nonlinear models with homophily (Golub and Jackson, 2012) or persuasion bias (DeMarzo et al., 2003). Around a fixed point, the Jacobian determines which disagreement dimensions propagate and how slowly they decay. If the local propagation matrix has the island structure, the relevant dimensions are group disagreements as in the island model; if it has clustered structure, coarse targeting responds locally to slow-moving cluster disagreements. Thus the mechanism is robust within the region where the linear approximation is accurate: coarse targeting prioritizes disagreement dimensions that decay slowly, even when all disagreement is ultimately transient.

S4 Computational algorithm

S4.1 Comparison of algorithms

In general, the optimal coarse targeting weights do not admit a closed-form solution, so computation relies on numerical methods that exploit the fixed-point characterization in equation (7). This appendix describes the numerical methods used to search for an optimal exposure rule. We consider two approaches. The first is a *nested fixed-point* algorithm directly implied by the first-order condition in (7), which evaluates the gradient efficiently by

solving the Lyapunov equation $(\sqrt{\delta}\tilde{A})Z(\sqrt{\delta}\tilde{A})' - Z + \mathbf{x}_0\mathbf{x}_0' = 0$. Related nested fixed-point algorithms are common in dynamic discrete choice problems (Rust, 1987). The second is a class of *projected gradient methods* (PGM) for constrained nonconvex optimization, which serve as a natural benchmark and are widely used in related settings (Gao et al., 2021; Solimine and Meyer-Baese, 2022).

Both approaches require repeated evaluation of the value function and its gradient. For a given $\mathbf{b}$, this involves solving an algebraic Riccati equation for the continuation value $K^*(\mathbf{b})$ and a discrete Lyapunov equation for the associated Gramian (Vaughan, 1970; Laub, 1979).

A. Common computational primitives. Both the nested fixed-point iteration and the projected gradient methods rely on the same underlying computational primitives. For any candidate target vector $\mathbf{b}$, the continuation value $K^*(\mathbf{b})$ is obtained as the unique stabilizing solution to an algebraic Riccati equation. Conditional on $K^*(\mathbf{b})$, the induced feedback vector $\mathbf{p}(\mathbf{b})$ and transition matrix $\tilde{A}(\mathbf{b}) = A - \mathbf{b}\mathbf{p}(\mathbf{b})'$ are computed, and the discounted Gramian associated with the induced opinion dynamics is obtained by solving a discrete Lyapunov equation. Together, these objects determine the gradient $\nabla_{\mathbf{b}}v(\mathbf{b})$ appearing on the right-hand side of the first-order condition (7).¹⁶

Algorithm 3 summarizes the steps required to compute $\nabla_{\mathbf{b}}v(\mathbf{b})$ for a given $\mathbf{b}$. Both classes of algorithms discussed below repeatedly call this routine; they differ only in how the target vector $\mathbf{b}$ is updated across iterations, not in the evaluation of the underlying matrix equations.

B. Projected gradient method (benchmark). As a benchmark, we implement a projected gradient method (PGM) for the constrained maximization of the value function $v(\mathbf{b})$ subject to the norm constraint $\|\mathbf{b}\| = 1$. Projected gradient methods are standard tools for constrained, nonconvex optimization problems and have been used in related targeting environments (Gao et al., 2021).

Starting from an initial exposure vector $\mathbf{b}_0$, the method updates $\mathbf{b}$ iteratively in the direction of the gradient $\nabla_{\mathbf{b}}v(\mathbf{b}_k)$, projected onto the tangent space of the constraint set. Concretely,

¹⁶The gradient formula is valid for any symmetric positive-definite payoff matrix Q . The derivation does not differentiate Q with respect to $\mathbf{b}$; Q enters through the Riccati solution $K^*(\mathbf{b})$. Thus the same algorithm applies to the population-weighted specification Q^{pop} .

each update takes the form

$$\mathbf{b}_{k+1} = \frac{\mathbf{b}_k + \eta_k(I - \mathbf{b}_k \mathbf{b}_k') \nabla_{\mathbf{b}} v(\mathbf{b}_k)}{\|\mathbf{b}_k + \eta_k(I - \mathbf{b}_k \mathbf{b}_k') \nabla_{\mathbf{b}} v(\mathbf{b}_k)\|},$$

where $(I - \mathbf{b}_k \mathbf{b}_k')$ removes the component of the gradient that points in the direction of $\mathbf{b}_k$, and η_k is a step size.

We use an adaptive step size: η_k is selected by a line search along the projected-gradient direction to maximize the value function locally. Fixed-step implementations are sensitive to the selected step size, so the reported comparison uses adaptive PGM. Algorithm 1 summarizes the procedure. Under standard regularity conditions, PGM converges to a stationary point of the constrained problem, though convergence is generally local and may depend on initialization.

C. Nested fixed-point iteration. The second approach is a nested fixed-point iteration. Rather than updating $\mathbf{b}$ along a gradient direction with an explicit step size, this method constructs a sequence $\{\mathbf{b}_k\}$ by repeatedly evaluating the fixed-point mapping implied by the optimality condition.

Starting from an initial exposure vector $\mathbf{b}_0$, each iteration updates $\mathbf{b}$ by plugging the current iterate into the right-hand side of the first-order condition and renormalizing to satisfy the norm constraint,

$$\mathbf{b}_{k+1} \propto \nabla_{\mathbf{b}} v(\mathbf{b}_k),$$

with proportionality chosen so that $\|\mathbf{b}_{k+1}\| = 1$. Algorithm 2 summarizes this update rule.

The method is *nested* because evaluation of the fixed-point mapping at $\mathbf{b}_k$ requires solving for the continuation value $K^*(\mathbf{b}_k)$ and the associated discounted Gramian, as described in Part A. In contrast to the projected gradient method, the nested fixed-point iteration does not require choosing a step size or performing a line search; the update is entirely determined by the fixed-point characterization of optimal targeting. For a given iterate $\mathbf{b}_k$, the matrix equation defining $K^*(\mathbf{b}_k)$ is solved directly using a standard Schur decomposition–based algorithm for algebraic Riccati equations (Laub, 1979). This avoids iterative solution of the inner fixed point and allows each outer iteration to be computed reliably using well-established numerical linear algebra routines.

Algorithm 1: Adaptive PGM

Data: $A, \mathbf{x}_0, \delta, c, Q$ **Result:** Stationary candidate exposure vector $\hat{\mathbf{b}}$ $\mathbf{b}_0 \leftarrow \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}$ if $A\mathbf{x}_0 \neq \mathbf{0}$; otherwise $\mathbf{b}_0 \leftarrow \mathbf{e}_1 \quad k \leftarrow 0;$ **while not converged do** $\mathbf{g}_k \leftarrow \nabla_{\mathbf{b}} v(\mathbf{b}_k);$ $\mathbf{g}_k^\perp \leftarrow (I - \frac{\mathbf{b}_k \mathbf{b}_k'}{\mathbf{b}_k' \mathbf{b}_k}) \mathbf{g}_k;$ Define $\mathbf{f}_k(\eta) = \frac{\mathbf{b}_k + \eta \mathbf{g}_k^\perp}{\|\mathbf{b}_k + \eta \mathbf{g}_k^\perp\|};$ $\eta_k^* \leftarrow \arg \max_{\eta} v(\mathbf{f}_k(\eta));$ $\mathbf{b}_{k+1} \leftarrow \mathbf{f}_k(\eta_k^*);$ $k \leftarrow k + 1;$ **end****return** $\mathbf{b}_k;$

Algorithm 2: Nested fixed-point

Data: $A, \mathbf{x}_0, \delta, c, Q$ **Result:** Stationary candidate exposure vector $\hat{\mathbf{b}}$ $\mathbf{b}_0 \leftarrow \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}$ if $A\mathbf{x}_0 \neq \mathbf{0}$; otherwise $\mathbf{b}_0 \leftarrow \mathbf{e}_1 \quad k \leftarrow 0;$ **while not converged do** $\mathbf{b}_{k+1} \leftarrow \nabla_{\mathbf{b}} v(\mathbf{b}_k);$ $\mathbf{b}_{k+1} \leftarrow \mathbf{b}_{k+1} / \|\mathbf{b}_{k+1}\|;$ $k \leftarrow k + 1;$ **end****return** $\mathbf{b}_k;$

Algorithm 3: Compute gradient $\nabla_{\mathbf{b}} v(\mathbf{b})$

Data: $\mathbf{b}, A, \delta, c, Q$ **Result:** Gradient $\nabla_{\mathbf{b}} v(\mathbf{b})$ Solve $K^* = Q + \delta A' K^* (I + \frac{\delta}{c} \mathbf{b} \mathbf{b}' K^*)^{-1} A$ for $K^*;$ $\mathbf{p}' \leftarrow (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \delta \mathbf{b}' K^* A;$ $\tilde{A} \leftarrow A - \mathbf{b} \mathbf{p}';$ Solve $\sqrt{\delta} \tilde{A} Z \sqrt{\delta} \tilde{A}' - Z = -\mathbf{x}_0 \mathbf{x}_0'$ for $Z;$ **return** $2\delta K^* \tilde{A} Z \mathbf{p};$

D. Practical implementation details. Both algorithms require an initial exposure vector and a stopping criterion. Before comparing successive directions, we choose the sign so that $\mathbf{b}_{k+1}' \mathbf{b}_k \geq 0$. In the algorithm-comparison simulations, iterations terminate when the sign-aligned iterate distance satisfies $\|\mathbf{b}_{k+1} - \mathbf{b}_k\|_1 < 10^{-6}$ or after 200 iterations. In the empirical adaptive-PGM implementation, the projected-gradient tolerance is 10^{-9} , the line-search tolerance is 10^{-7} , the objective-improvement tolerance is 10^{-10} , and the cap is 60 outer iterations. The multi-start initializations used for the application are documented in Appendix S5.2. The fixed-point map is fast when stable: it converges in the opening island example, the $Q = I$ application, and most simulation instances. The population-weighted empirical application instead uses adaptive projected gradient, because the undamped fixed-

point iteration frequently cycles from both informed and random starts when $Q = Q^{\text{pop}}$.

E. Simulation design. To evaluate the numerical performance of the algorithms described above, we conduct simulations on synthetic networks of varying size and structure. The goal of these experiments is not to assess economic outcomes, but to compare convergence behavior and computational cost across methods under controlled conditions.

We use two simulation panels. The algorithm-comparison panel uses $n \in \{2, 4, \dots, 32\}$ and 25 independent networks at each size. A separate scalability panel uses $n \in \{25, 50, 100, 150, 200, 300, 400, 600\}$ and 10 independent networks at each size. Adaptive PGM is run through $n = 200$, while the fixed-point method is run for every listed size. Each network is drawn from a uniform Dirichlet ensemble: each row of A is sampled independently from a Dirichlet distribution on the probability simplex, ensuring that A is row-stochastic and dense. This design produces asymmetric networks with heterogeneous influence patterns, similar in spirit to the empirical Facebook network used in the application.

For each network, we draw an initial opinion vector $\mathbf{x}_0$ as follows:

$$\mu_s \sim \mathcal{N}(0, 5), \quad \sigma_s \sim \text{Uniform}(0, 5), \quad \mathbf{x}_0 \sim \mathcal{N}(\mu_s \mathbf{1}, \sigma_s^2 I).$$

This construction generates both low- and high-dispersion initial conditions, allowing us to assess algorithm performance across a range of disagreement profiles.

F. Results. Figure 5 summarizes the comparison panel. It reports wall time and the number of iterations required for convergence as a function of network size, averaged across simulation draws. In the comparison panel, nested fixed-point averages 4.13 outer iterations and 0.018 seconds per completed instance (0.0045 seconds per iteration)¹⁷. Adaptive PGM averages fewer iterations (2.26) but 0.49 seconds per completed instance (0.217 seconds per iteration), due to the expensive line search operation. The scalability panel shows the same efficiency ranking. The fixed-point method averages approximately four iterations through $n = 600$ and 5.1 iterations at $n = 1000$. At $n = 200$, it converges in 34 seconds on average, compared with 1,032 seconds for the adaptive PGM. Each iteration of both methods involves solving matrix equations with $\mathcal{O}(n^3)$ complexity. Adaptive PGM additionally incurs repeated value-function evaluations during line search, leading to a per-iteration cost of

¹⁷Simulations were run on a 13th Gen Intel Core i7-1355U at 1700 Mhz.

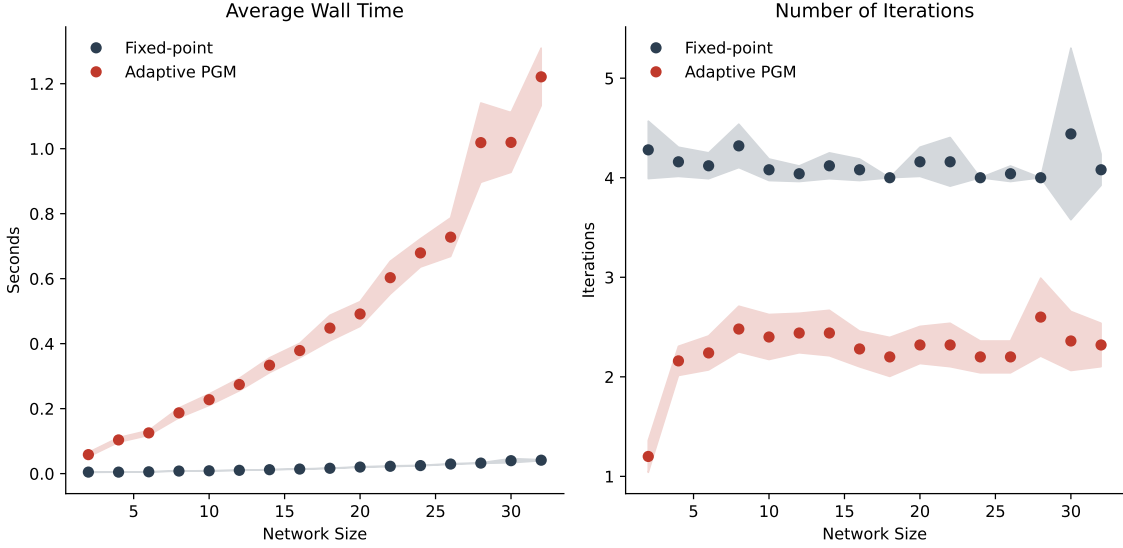


Figure 5: Convergence speed in the algorithm-comparison panel ($n \leq 32$)

order $\mathcal{O}(\log(1/\epsilon) n^3)$, where ϵ denotes the line-search tolerance.¹⁸ This difference reflects the fact that the fixed-point update directly implements the first-order condition, whereas gradient-based methods perform a sequence of local ascent and line-search steps.

When the fixed-point mapping is unstable, adaptive projected gradient provides the robust fallback. In simulation instances where both methods converge, their returned directions agree within the stopping tolerance. The comparison therefore highlights a trade-off between speed and robustness rather than differences in optimality.

S5 Data appendix

S5.1 Proof and construction details for Section 6.1

The model presented throughout the paper takes the opinion propagation matrix A as a primitive, but the construction of the empirical object from the available data is not trivial. We develop a construction that is defensible under a natural microfoundation, connecting individual-level social interactions to state-level opinion dynamics. The argument has two steps: first, we derive the state-level form of A from an individual population-level

¹⁸Parallel or approximate solvers (Laine and Tomlin, 2019) may reduce this gap for large or sparse networks. Each reported instance is solved serially; independent simulation draws may be distributed across processors.

model. Second, we show that the unknown scale of the SCI matrix is cancelled through row-normalization, so that A is identified directly from the observable aggregate data.

We assume that individual initial opinions are distributed around the state-level average

$$y_{0,i} = x_{0,s(i)} + \epsilon_i,$$

where the shocks ϵ_i are independent across individuals and social ties, mean zero, and have finite variance σ_ϵ^2 . Opinions follow a DeGroot updating process at the individual level: $y_{t+1,i} = \sum_j T_{ij}^{\text{ind}} y_{t,j}$, where $T_{ij}^{\text{ind}} = A_{ij}^{\text{ind}}/d_i^{\text{ind}}$ is the row-stochastic normalization of the individual adjacency matrix.

Social ties form according to a state-level stochastic block model: conditional on state memberships, $A_{ij}^{\text{ind}} \sim \text{Bernoulli}(p_{s(i),s(j)})$ independently across pairs. The cross-state probability matrix $\mathbf{P} = [p_{ss'}]$ is the structural object that summarizes the model. The next result gives a fixed-date large-population justification for the state-level recursion.

Proposition S2. *Fix the number of states. Let $N = \sum_s N_s$, $N_{\min} = \min_s N_s$, and suppose $N_{\min}/N$ is bounded away from zero. Suppose also that there is a constant $\underline{\mu} > 0$ such that every expected individual degree satisfies $\mu_s \geq \underline{\mu}N$. Let*

$$x_{t,s}^{(N)} = \frac{1}{N_s} \sum_{i:s(i)=s} y_{t,i}$$

denote the realized state-average opinion. Define

$$\hat{A}_{ss'}^{(N)} = \frac{p_{ss'} N_{s'}}{\mu_s}, \quad \mu_s = \sum_{s''=1}^n p_{ss''} N_{s''}, \quad (\text{S9})$$

and let $\bar{\mathbf{x}}_0 = (x_{0,1}, \dots, x_{0,n})'$. For every fixed date t ,

$$\max_s \left| x_{t,s}^{(N)} - \left((\hat{A}^{(N)})^t \bar{\mathbf{x}}_0 \right)_s \right| \xrightarrow{p} 0 \quad \text{as } N_{\min} \rightarrow \infty.$$

Moreover, $\hat{A}^{(N)}$ is row-stochastic.

Proof of Proposition S2: Let $R^{(N)}$ be the state-averaging matrix, let $P^{(N)} = \mathbb{E}[A^{\text{ind}}]$, and

define $\bar{D}^{(N)} = \text{diag}(\mu_{s(1)}, \dots, \mu_{s(N)})$ and $\bar{T}^{(N)} = (\bar{D}^{(N)})^{-1} P^{(N)}$. Thus

$$\bar{T}_{ij}^{(N)} = \frac{p_{ss'}}{\mu_s} \quad \text{when } s(i) = s, s(j) = s', \quad R^{(N)} \bar{T}^{(N)} = \hat{A}^{(N)} R^{(N)}.$$

Applying the rectangular Matrix Bernstein inequality to $A^{\text{ind}} - P^{(N)}$ and the scalar Bernstein inequality to the individual degrees gives

$$\|T^{\text{ind}} - \bar{T}^{(N)}\|_2 = O_p\left(\sqrt{\frac{\log N}{N}}\right); \quad (\text{S10})$$

see Tropp (2012, Theorem 1.6) and the related random-graph concentration results in Chung and Radcliffe (2011); Le et al. (2017). The entries of $\bar{T}^{(N)}$ are at most $(\underline{\mu}N)^{-1}$, so $\|\bar{T}^{(N)}\|_2 \leq \underline{\mu}^{-1}$. Equation (S10) then also gives $\|T^{\text{ind}}\|_2 = O_p(1)$. For the fixed date t , the telescoping identity

$$(T^{\text{ind}})^t - (\bar{T}^{(N)})^t = \sum_{\ell=0}^{t-1} (T^{\text{ind}})^\ell (T^{\text{ind}} - \bar{T}^{(N)}) (\bar{T}^{(N)})^{t-1-\ell}.$$

It follows that

$$\|(T^{\text{ind}})^t - (\bar{T}^{(N)})^t\|_2 = O_p\left(\sqrt{\frac{\log N}{N}}\right) = o_p(1).$$

Writing the initial individual-opinion vector as $\mathbf{y}_0 = L^{(N)} \bar{\mathbf{x}}_0 + \boldsymbol{\epsilon}$, where $L^{(N)}$ replicates each state mean across its residents, gives

$$\mathbf{x}_t^{(N)} - (\hat{A}^{(N)})^t \bar{\mathbf{x}}_0 = R^{(N)} [(T^{\text{ind}})^t - (\bar{T}^{(N)})^t] \mathbf{y}_0 + (\hat{A}^{(N)})^t R^{(N)} \boldsymbol{\epsilon}.$$

Now $\|R^{(N)}\|_2 = N_{\min}^{-1/2}$ and $\|\mathbf{y}_0\|_2 = O_p(\sqrt{N})$, so the first term is $o_p(1)$. The s th component of $R^{(N)} \boldsymbol{\epsilon}$ has variance σ_ϵ^2 / N_s , and hence

$$\|R^{(N)} \boldsymbol{\epsilon}\|_2 = O_p(N_{\min}^{-1/2}) = o_p(1).$$

Since $(\hat{A}^{(N)})^t$ is row-stochastic and n is fixed, its operator norm is bounded, so the second term is also $o_p(1)$. Finally,

$$\sum_{s'} \hat{A}_{ss'}^{(N)} = \frac{\sum_{s'} p_{ss'} N_{s'}}{\mu_s} = 1,$$

which proves row-stochasticity. $\square$

Remark S1. Proposition S2 is an asymptotic, fixed-date result and does not provide a finite-sample error constant. Large state populations make state-average idiosyncratic-opinion noise small, but the theorem alone does not quantify the exact magnitude of the approximation error of the aggregated propagation matrix in a finite sample.

Remark S2. The construction does not require $\hat{A}$ and $\mathbf{x}_0$ to be independent. For example, states may have latent characteristics $\mathbf{v}_s$ that jointly determine average opinions, $x_{0,s} = f_0(\mathbf{v}_s) + \epsilon_s$, and link propensities, $p_{ss'} = W(\mathbf{v}_s, \mathbf{v}_{s'})$. Proposition S2 conditions on these state characteristics, so correlation between initial opinions and network connectivity does not alter the aggregation argument.

Proposition S2 shows that the aggregate opinion dynamics can be approximated by the state-level cross-state probability matrix $\mathbf{P} = [p_{ss'}]$, the core object of relevance to the model. Due to privacy restrictions, however, the SCI provides a signal proportional to this link probability. The following result shows that the unknown scale does not affect the propagation matrix.

Lemma S1. *Suppose $SCI_{ss'} = \kappa \cdot p_{ss'}$ for an unknown scalar $\kappa > 0$ that is common across state pairs. Then $\hat{A}^{(N)}$ defined in (S9) satisfies*

$$\hat{A}_{ss'}^{(N)} = \frac{SCI_{ss'} N_{s'}}{\sum_{s''=1}^n SCI_{ss''} N_{s''}},$$

so κ cancels identically in the row-normalization. Therefore $\hat{A}^{(N)}$ is point-identified from the observed SCI and state populations, regardless of the scale factor.

Proof of Lemma S1: Since $p_{ss'} = SCI_{ss'}/\kappa$, substituting into (S9) gives

$$\hat{A}_{ss'}^{(N)} = \frac{(SCI_{ss'}/\kappa) N_{s'}}{\sum_{s''} (SCI_{ss''}/\kappa) N_{s''}} = \frac{SCI_{ss'} N_{s'}}{\sum_{s''} SCI_{ss''} N_{s''}}. \quad (\text{S11})$$

The common factor $1/\kappa$ cancels. $\square$

Taken together, Proposition S2 and Lemma S1 provide a fixed-date large-population justification for using the row-normalized, population-weighted SCI matrix as the empirical

counterpart of the model primitive A . Following Bailey et al. (2021), we aggregate county-level SCI to states according to

$$\text{SCI}_{ab} = \sum_{i \in C(a), j \in C(b)} \frac{\text{Pop}_i}{\text{Pop}_a} \frac{\text{Pop}_j}{\text{Pop}_b} \text{SCI}_{ij}.$$

This population-weighted average preserves proportionality to the state-pair link probability with the same scale κ . We then multiply each destination column by its state population and row-normalize as in Lemma S1. The resulting matrix is generally asymmetric because destination populations and row-normalization denominators differ across states, even though the underlying SCI is symmetric. We retain self-links, which are removed from Figure 2 for visualization clarity.

Opinions are county `consensusOppose` estimates from the 2023 Yale Climate Opinion Maps (Howe et al., 2015; Marlon et al., 2022), aggregated to the 50 states and the District of Columbia using county population shares. The network uses 2020 county populations from `co-est2020.csv`; Q^{Pop} and the opinion aggregation use the corresponding 2020 population vintage in the replication code. Jurisdictions are ordered alphabetically by postal abbreviation, including DC. All state pairs are observed, no row has zero mass, self-links are retained in $\hat{A}$, and each row is normalized to one, justified by Proposition S2. As a specification check, we replace the county-aggregated opinions with the published state estimates in the same Yale workbook. The California–Texas exposure gap falls from 0.0032 to 0.0009. Coarse capture remains 99.91%, effective dimension is 1.03, and the leading share is 98.68%. The workbook does not provide state-specific standard errors or a covariance matrix, so we treat this as a specification sensitivity rather than formal statistical inference. The replication script records the source files and regenerates the reported tables and figures.

S5.2 Computation and robustness

We compute the reported population-weighted coarse candidate using the adaptive projected-gradient method in Algorithm 1. The multi-start search uses six informed initializations, $Q\hat{A}\mathbf{x}_0$, $\hat{A}\mathbf{x}_0$, $\mathbf{x}_0$, authority, hub, and the uniform direction, together with random seeds $0, \dots, 5$. All twelve starts reach the same objective to ten decimal places. A separate sphere-constrained optimization changes the exposure direction by less than 1.5×10^{-6} and leaves all reported values unchanged at the displayed precision. The undamped fixed-point

method converges in the opening example and the $Q = I$ application but cycles under Q^{pop} , so it is not used for the primary empirical results. We retain the highest-value candidate and compare it with every reported fixed heuristic. Full targeting uses the unrestricted vector intervention $\mathbf{u}_t$ defined in Section 2. For any payoff matrix Q , the no-intervention loss matrix solves $K^\varnothing = Q + \delta A' K^\varnothing A$.

The primary specification uses $Q = Q^{\text{pop}}$. Authority and hub centrality are the leading right and left singular vectors of $\hat{A}$. One period denotes one application of the state updating matrix, not a calibrated calendar interval. The parameters δ and c are chosen to be illustrative (rather than calibrated), motivating the reported parameter grid. Because $\|\mathbf{b}\|_2 = 1$, $\|\mathbf{u}_t\|_2 = |r_t|$, so c prices squared Euclidean intervention magnitude relative to the normalized opinion-loss scale. Under the baseline sign normalization, all coordinates of the reported exposure vector are positive and the optimal intensity does not reverse sign along the realized path. For completeness, Table 4 reports the wider cost grid and the additional fixed heuristics omitted from the main text.

Table 4: Policy comparison across intervention costs ($\delta = 0.9$, $Q = Q^{\text{pop}}$). Entries report the share of attainable gains captured. Entries displayed as 100.0% are rounded.

	$c = 1$	$c = 10$	$c = 100$	$c = 1000$	$c = 10000$
Opinion ($\mathbf{b} \propto \mathbf{x}_0$)	98.3%	91.1%	72.0%	51.3%	45.2%
Network-updated ($\mathbf{b} \propto \hat{A}\mathbf{x}_0$)	98.5%	91.7%	73.3%	53.0%	46.9%
Payoff-updated ($\mathbf{b} \propto Q^{\text{pop}}\hat{A}\mathbf{x}_0$)	94.3%	97.8%	99.4%	99.4%	99.3%
Authority ($\mathbf{b} \propto \mathbf{v}_1$)	97.2%	99.1%	97.8%	94.2%	92.6%
One-period direction	99.6%	99.2%	99.5%	99.4%	99.3%
Period-by-period myopic retargeting	99.7%	91.5%	52.2%	25.1%	20.2%
Coarse ($\mathbf{b} = \hat{\mathbf{b}}$)	99.9%	99.9%	100.0%	100.0%	100.0%

Table 5 reports the largest state-level differences between the reusable and one-period exposure directions at the baseline.

Table 6 reports robustness to discount factors away from the baseline $\delta = 0.9$; Table 4 reports the baseline cost grid in greater detail. The last column uses a separate raw-value normalization, $|v^c - v^f|/|v^f|$, presenting the value gap as a proportion of full targeting value. Through this lens, the coarse rule stays within 0.71% of full targeting under all tested parameter values. The capture columns show that the computed coarse rule dominates every reported fixed heuristic at every cell, while each individual heuristic has a regime where it

Table 5: Reusable and one-period exposure directions at the baseline ($\delta = 0.9$, $c = 10$, $Q = Q^{\text{pop}}$). Both directions have unit Euclidean norm. Absolute differences are computed from the unrounded exposure weights.

State	Reusable	One-period	Absolute difference
California	0.3280	0.4366	0.1086
Texas	0.3248	0.3983	0.0735
Mississippi	0.0921	0.0604	0.0317
Minnesota	0.1337	0.1025	0.0313
Alabama	0.1331	0.1022	0.0309

falls short. Table 7 reports population-weighted capture under recurrent shocks of size σ

Table 6: Population-weighted capture $(v(\mathbf{b}) - v^{\emptyset})/(v^f - v^{\emptyset})$ across (δ, c) ($Q = Q^{\text{pop}}$). Full targeting is the 100% benchmark; entries displayed as 100.0% are rounded. The last column is the coarse-to-full gap $|v^c - v^f|/|v^f|$.

δ	c	Coarse	Authority	Opinion	Hub	coarse/full gap
0.80	1	99.82%	95.39%	96.14%	97.60%	0.45%
0.80	10	99.87%	98.49%	81.64%	94.64%	0.13%
0.80	100	99.99%	95.80%	56.79%	82.33%	0.00%
0.95	1	99.95%	98.49%	99.20%	99.26%	0.65%
0.95	10	99.95%	99.48%	95.67%	98.80%	0.30%
0.95	100	100.0%	98.88%	84.48%	94.77%	0.01%
0.99	1	99.99%	99.68%	99.85%	99.84%	0.71%
0.99	10	99.99%	99.89%	99.16%	99.77%	0.36%
0.99	100	100.0%	99.78%	96.73%	98.96%	0.01%

at $\delta = 0.9$ and $c = 10$. For each σ , we recompute the coarse direction using the expected discounted opinion-path matrix, which adds the shock covariance to the deterministic path. Both fixed-direction rules lose out to full targeting as σ grows. The table is computed analytically from expected discounted losses under shock covariance $\sigma^2 I$; it does not use Monte Carlo simulation.

S5.3 Effective dimension of the opinion path

Table 8 reports the concentration of the population-weighted realized path $Z(\hat{\mathbf{b}})$ across the parameter grid, underlying the effective-dimension discussion in Section 6.3. The effective

Table 7: Population-weighted capture under recurrent shocks of size σ (with $\delta = 0.9$, $c = 10$, $Q = Q^{\text{pop}}$). Full targeting is the 100% benchmark at every σ .

σ	0	0.1	0.2	0.3	0.5	1.0
Coarse ($\hat{\mathbf{b}}$)	99.9%	97.2%	91.3%	85.6%	78.2%	72.4%
Authority ($\mathbf{v}_1$)	99.1%	96.5%	90.8%	85.1%	77.7%	71.6%

path dimension stays near one throughout, so the path remains low-dimensional regardless of (δ, c) . Let $\bar{\phi} = \text{tr}(\Phi)/51$. The penultimate column reports the intervention error on the dominant unit eigenvector $v_{\max}$ of Z relative to this mean, $v'_{\max} \Phi v_{\max} / \bar{\phi}$. The final column reports the realized coarse-to-full loss as a share of attainable gains.

Table 8: Concentration of the population-weighted realized opinion path $Z(\hat{\mathbf{b}})$, network of the 50 states and the District of Columbia ($Q = Q^{\text{pop}}$). “Lead share” is the largest eigenvalue of Z over its trace; $d_{\text{eff}} = (\sum_k \sigma_k)^2 / \sum_k \sigma_k^2$; $v'_{\max} \Phi v_{\max} / \bar{\phi}$ is the intervention error on Z ’s leading direction relative to its mean.

δ	c	Z lead share	d_{eff}	$v'_{\max} \Phi v_{\max} / \bar{\phi}$	Δ/gain
0.80	1	96.72%	1.07	9.0×10^{-4}	0.185%
0.80	10	99.02%	1.02	1.8×10^{-4}	0.127%
0.80	100	99.79%	1.00	6.5×10^{-7}	0.007%
0.90	1	96.14%	1.08	1.3×10^{-3}	0.101%
0.90	10	98.59%	1.03	3.2×10^{-4}	0.087%
0.90	100	99.75%	1.01	2.0×10^{-6}	0.007%
0.95	1	95.86%	1.09	1.5×10^{-3}	0.052%
0.95	10	98.34%	1.03	4.1×10^{-4}	0.050%
0.95	100	99.69%	1.01	4.7×10^{-6}	0.005%
0.99	1	95.63%	1.09	1.7×10^{-3}	0.011%
0.99	10	98.11%	1.04	5.1×10^{-4}	0.011%
0.99	100	99.61%	1.01	1.0×10^{-5}	0.001%